NSD-ISS Stage Prediction with Calibrated Uncertainty: A Benchmarked Machine Learning Framework for Biological Staging in Parkinson's Disease

Blair Dupre

The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) redefines Parkinson's disease (PD) through biological anchors, but no computational model predicts NSD-ISS stage from routinely collected clinical data. Using PPMI (n=2,201), we benchmark a strict-circularity specification excluding every variable entering the NSD-ISS staging algorithm—putamen striatal-binding ratio, MDS-UPDRS-III total, NP1 cognition, and the caudate/putamen ratio (which inherits the D-anchor signal through its denominator)—retaining 21 non-staging features. CatBoost (default) achieves binary NSD+ AUC 0.901 [95% CI 0.887–0.915] and NSD+ sub-staging AUC 0.908 [0.889–0.925] on 5-fold stratified cross-validation; five SOTA methods on the identical primary (CatBoost default/HPO, LightGBM HPO, TabPFN v2, AutoGluon) are statistically equivalent at ε=0.02 (paired-bootstrap TOST: 34/40 pairs). A paired-bootstrap ablation against a 12-feature clinical-only subset reduces binary AUC by -17.6 percentage points ($\Delta = -0.176$, $p<0.0001$) but leaves NSD+ sub-staging statistically unchanged ($\Delta = +0.008$, p=0.21), validating a two-stage deployment (biological confirmation once, then clinical-only sub-staging) and reinforced by a caudate-vs-putamen residualization sensitivity (binary -10.7 pp; NSD+ unchanged). Cross-conformal CV+ achieves >90% internal marginal coverage with near-singleton sets ($|C|$ 0.96–1.27); external coverage on BioFIND (n=103) holds at 0.915 for binary but collapses to 0.499 (three-class) and 0.707 (NSD+). Per-target temperature scaling reduces binary expected calibration error from 0.053 to 0.020 with conformal coverage preserved. External binary balanced accuracy 0.516 reflects PPMI's NSD-negative class mixing healthy controls and prodromals; PD-only retraining mitigates this at internal-AUC cost (0.738→0.677). Five pre-registered confounder analyses (age, sex, protocol-LOCO, site-LOSO) find no evidence of confounding. This is the first strict-circularity benchmark for NSD-ISS biological staging with calibrated uncertainty and deployment-realistic external-calibration assessment.

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I. Introduction

Parkinson's disease (PD) has traditionally been staged using clinical severity scales such as the Movement Disorder Society-sponsored Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [23] and the MDS clinical diagnostic criteria [35]. These scales describe downstream motor manifestations of pathology that has typically been developing for years: pathological staging of Lewy pathology was proposed by Braak *et al.* [6], long before cerebrospinal-fluid alpha-synuclein seed amplification assays and dopamine-transporter (DaT) SPECT made *in vivo* biological staging operational at scale in cohorts such as the Parkinson's Progression Markers Initiative (PPMI) [30]. In January 2024, Simuni *et al.* [48] proposed the Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS), redefining PD through two biological anchors: neuronal alpha-synuclein (S, assessed by seed amplification assay, SAA) and dopaminergic dysfunction (D, assessed by DaT-SPECT imaging). The NSD-ISS assigns patients to stages 0 through 6, creating a biologically-grounded framework spanning preclinical through severe disability, and has been validated longitudinally: biological stage predicts time-to-clinical-milestones [49], staging replicates across cohorts including BioFIND [40], and 5-year stage transitions show consistent median times of 1.2 years (2B to 3) and 5.0 years (3 to 4) [50].

The NSD-ISS framework has also attracted principled critiques. Espay *et al.* [18] showed that dopaminergic medication status confounds several of the clinical sub-staging variables, leading to the recommendation that staging models be trained on cohorts with explicit medication-status stratification—a constraint we adopt in the present work and discuss further in §V.

Despite rapid clinical adoption, the NSD-ISS literature remains predominantly descriptive—no machine-learning model has been reported that predicts NSD-ISS biological stage from routinely-collected PPMI data with calibrated uncertainty, and the related PD-ML literature [1], [13], [38] has not targeted the NSD-ISS framework directly. Reporting guidance for such clinical prediction models is provided by the TRIPOD+AI statement [10]. The primary research question of this study is therefore: *can a benchmarked machine-learning framework with calibrated uncertainty predict NSD-ISS biological stage from routinely-collected PPMI clinical and imaging data,*

TABLE I
FOUR NSD-ISS TARGET FORMULATIONS EVALUATED IN THIS STUDY.

Target	K	What it predicts
Binary	2	NSD+ (stages 1+) vs. NSD− (stage 0)
Three-class	3	Early (0–1) vs. mild (2B) vs. impaired (3–4)
Full ordinal	5	All stages: 0, 1, 2B, 3, 4
NSD-positive	4	NSD+ sub-staging: 1, 2B, 3, 4

including cross-cohort transportability and a clinical-only feature subset for sites lacking DaT-SPECT or SAA access? Five specific objectives operationalise this question:

1) Benchmark seven tabular and one graph-based model for NSD-ISS stage prediction across four target formulations using PPMI data (n=2,201, §III).
2) Compare tabular (gradient-boosted trees) versus graph-based (graph attention network (GAT) with patient-similarity graphs) approaches on an identical feature set.
3) Quantify the contribution of biological markers versus clinical-only features, using a feature-ablation protocol matched to site-level resource availability.
4) Evaluate model transportability through external validation on BioFIND (n=118, with NSD-ISS ground truth) and prediction-only application to PDBP (n=893), using a pre-specified handling protocol for the training-label confound introduced by healthy-control inclusion in PPMI.
5) Calibrate model uncertainty through cross-conformal prediction (CV+) with distribution-free coverage guarantees, so that predictions are reported as small sets of plausible labels when individual-patient confidence is low.

The four NSD-ISS target formulations used throughout this paper span the full clinical-utility spectrum—binary detection of biological disease (screening), three-class triage (treatment routing), full ordinal staging (longitudinal monitoring), and NSD-positive sub-staging (trial enrichment at sites without imaging)—and are summarised in Table I.

II. Related Work

Machine learning for Parkinson's disease has focused primarily on clinical outcome prediction and diagnostic classification rather than biological staging. AdaMedGraph [1] combined APPNP graph propagation with AdaBoost (SAMME) for PD progression prediction using PPMI data. We reproduce AdaMedGraph in our benchmark (Table III ("Graph-based" rows)) and report its head-to-head performance on the NSD-ISS biological-staging targets; the per-feature-graph ensemble is competitive on binary classification but underperforms tree baselines on multi-class targets, consistent with the broader Grinsztajn 2022 finding. Recent work by Diaz-Rincon *et al.* [13] applied conformal prediction to PD medication needs assessment, establishing distribution-free coverage guarantees in a clinical PD context. However, neither of these approaches—nor any published model to date—targets the NSD-ISS biological staging framework introduced by Simuni *et al.* [48].

The choice of model architecture for clinical prediction tasks remains an active area of investigation. The dominant narrative, established by Grinsztajn *et al.* [26] and Shwartz-Ziv and Armon [47], is that gradient-boosted tree ensembles outperform deep learning on medium-sized tabular datasets. However, more recent work suggests this gap closes at smaller sample sizes: Zabërgja *et al.* [56] report deep-learning methods winning 31 of 34 benchmark tasks at $n < 5{,}000$, and the closer-look evaluation of Ye *et al.* [55] reaches the same conclusion for modern attention- and transformer-based tabular architectures [25]. Foundation-model tabular approaches have sharpened this shift: Hollmann *et al.* [59] demonstrate that TabPFN v2—a context-based transformer trained on synthetic tabular data—exceeds tuned CatBoost by an average $+0.13$ AUC across 150 benchmark datasets without any hyperparameter tuning. AutoML ensembling provides a complementary route: AutoGluon [60] achieves consistent leaderboard wins through weighted stacking across a heterogeneous model pool. These developments motivate our four-way tabular-SOTA comparison (CatBoost, LightGBM, TabPFN, AutoGluon) in §IV, which tests whether any single architecture dominates on NSD-ISS biological staging or whether—as Zabërgja, Hollmann, and Erickson collectively predict—tree boosters, foundation-model tabular, and AutoML ensembling converge at the $n \approx 2{,}000$ clinical-cohort regime where NSD-ISS is defined.

Conformal prediction [53] has emerged as a principled framework for uncertainty quantification in clinical ML, providing finite-sample marginal coverage guarantees without distributional assumptions. Huang *et al.* [27] extended conformal methods to graph neural networks, demonstrating that graph structure can be exploited for tighter prediction sets. Our work applies cross-conformal (CV+) prediction to NSD-ISS staging, combining the distribution-free guarantees of conformal inference with the clinical interpretability of set-valued predictions.

III. Methods

A. Study Design

This is a retrospective prediction-model development and external-evaluation study using data from four observational PD cohorts, reported following the TRIPOD+AI guidelines [10]. The per-cohort participant flow—enrollment, staging eligibility, and final role assignment—is summarised in Fig. 1.

B. Data Sources

PPMI (n=2,201): A prospective longitudinal study enrolling de novo PD patients, prodromal PD, and healthy controls across 33 sites. AMP-PD v4 data release.

BioFIND (n=118 PD): Cross-sectional biomarker study, moderate-to-advanced PD (diagnosed >5 years), 16 US sites. SAA data from three independent laboratories via LONI IDA.

PDBP (n=893 PD): NIH-funded longitudinal biomarker cohort. AMP-PD v4 clinical data.

HBS (n=649 PD): Single-center longitudinal study at Massachusetts General Hospital. AMP-PD v4 clinical data.

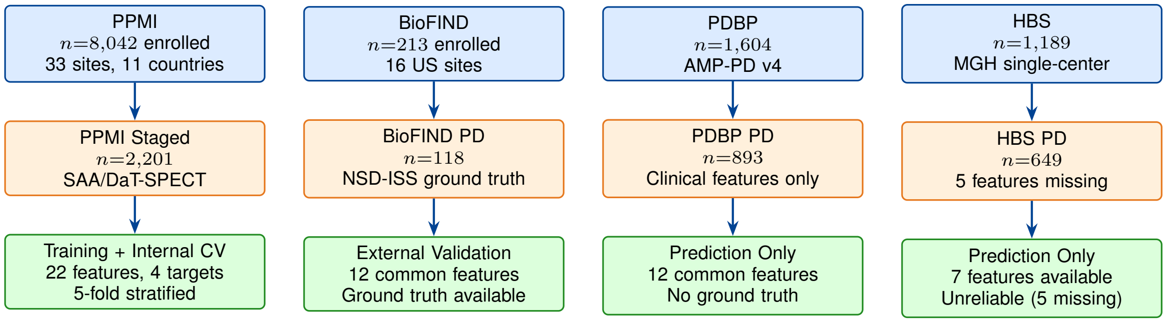

Fig. 1. **Study participant flow across the four PD cohorts.** PPMI served as the primary training cohort with NSD-ISS ground truth computed from SAA and DaT-SPECT biomarkers. BioFIND provided external validation with NSD-ISS staging replicated from Russo *et al.* [40]. PDBP and HBS were prediction-only cohorts without biological staging data.

C. NSD-ISS Biological Staging

NSD-ISS stages were computed from three anchors:

S anchor (synuclein): Seed amplification assay (SAA) positivity. Coverage: 12.6% of PPMI (277/2,201).

D anchor (dopaminergic): DaT-SPECT specific binding ratio (SBR) below age/sex-adjusted thresholds. Coverage: 97.1% (2,137/2,201).

Clinical staging: Among S+ and/or D+ patients, functional impairment assessed via MDS-UPDRS Part II (ADL) and MoCA (cognitive). Stage assignments followed Simuni *et al.* [48].

D. Feature Engineering

Feature-set hierarchy: Paper 1's primary feature specification is the strict-circularity 21-feature set, chosen so that no variable entering the Simuni 2024 NSD-ISS staging algorithm (either as a direct threshold input or as a component of a derivative feature) remains in the predictor vector. The 21 features span seven clinical domains (Table II). For external validation where not all features are available across cohorts, we use a 12-feature common subset (AGE_AT_BASELINE, SEX, UPDRS1_TOTAL, UPDRS2_TOTAL, UPDRS3_TREMOR, UPDRS3_RIGIDITY, UPDRS3_BRADYKINESIA, UPDRS3_AXIAL, UPDRS4_TOTAL, MOCA_TOTAL, ESS_TOTAL, RBD_TOTAL), which is the intersection of features present in all four cohorts (PPMI, BioFIND, PDBP, HBS). A pre-registered 33-feature sensitivity variant adding 11 further observed PPMI features (6 cortical thickness, 4 CSF biomarkers, 1 polygenic risk score) is evaluated in two ways: against the Multimodal GAT to test whether the GAT-vs-tree gap is architectural or feature-count-limited (Supplementary S-3, 3-modality 32-feature variant), and against the tabular models to test whether the canonical schema is under-powered (Supplementary S-4, pre-registered null result—the extension does not improve tabular performance and the halt rule fired). Table III reports the 22-feature reference specification alongside the 21-feature primary (the two specifications differ only in the inclusion/exclusion of the caudate/putamen ratio; all other features are shared); the strict-exclusion specification is the primary throughout §IV and §V.

Feature retention rationale: The 21-feature set retains one score per clinical domain (e.g., SCOPA-AUT total rather than the five subscales), dropping redundant subscale decompositions that share $\geq$ 80% variance with their totals. Caudate-based DaT-SPECT features are retained (left, right, mean, and asymmetry) because the caudate is a prognostically informative and staging-algorithm-independent striatal subregion; the putamen SBR and any derivative that depends on it are excluded per the circularity audit below.

Circularity audit (five exclusions): The NSD-ISS framework is itself rule-based: every patient's stage label is produced by thresholding a specific set of clinical and imaging variables (Simuni *et al.* [48]). A predictor model trained on those same thresholding variables would recover the staging rule rather than extract biological signal from them. To avoid this tautology we exclude five features from the 21-feature primary set:

- **PUTAMEN_L_SBR and PUTAMEN_R_SBR** enter the D-anchor branch of the NSD-ISS staging algorithm directly (dopaminergic deficit is defined as putamen SBR below a population z-score threshold).
- **CAUDATE_PUTAMEN_RATIO** was initially retained on the reasoning that a ratio, not the putamen value itself, carries the biological information; however a pre-registered sensitivity analysis (§III-D) shows that its denominator preserves the D-anchor signal—refitting without the ratio reduces binary AUC by Δ=0.077, well above the 0.03 MATERIAL threshold, so the feature is excluded as a second-order D-anchor leak.
- **NP3TOT** (MDS-UPDRS Part III total) enters the Simuni 2024 clinical sub-staging thresholds directly (motor-examination branch).
- **NP1COG** (MDS-UPDRS Part I cognition anchor, item 1.1) is the cognitive-staging trigger variable in the Simuni algorithm.

We retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial) because these are clinically distinct biological indicators, none of which is individually used in the Simuni threshold. A subscale-sum sensitivity analysis confirms that their joint inclusion changes binary AUC by $<$0.01 relative to using only the total (reported in §V-E). A further label-variable ablation (Supplementary S-6) removing

UPDRS1_TOTAL and UPDRS2_TOTAL from the 21-feature primary reduces binary AUC by <0.003 across all four targets, confirming that the classifier is not rediscovering the Simuni threshold rules from UPDRS totals—the residual predictive signal comes from non-staging variables.

High-missingness handling: Two features in the 22-feature set, UPDRS4_TOTAL (89.9% missing) and MOCA_TOTAL (83.5% missing), are dropped *universally across all model families* at model-fit time on the PPMI internal benchmark, including CatBoost. The reason is subtle and worth making explicit because it appears at first glance to contradict CatBoost's native missing-value support: CatBoost's split-finding routine encodes missing values as a separate category, which works well at moderate missingness rates ($\lesssim$ 30–50%) but at the $\gtrsim$ 80% rates of these two features effectively introduces a selection-effect signal rather than a clinical measurement (PPMI patients who skip MoCA or UPDRS-IV are systematically older, frailer, or earlier-stage; their "missing" bin is informative but reflects collection protocol, not biology). Dropping them yields the cleanest cross-model comparison and avoids attributing accuracy to a non-clinical artifact. For non-CatBoost models, where missing values are median-imputed within each training fold, the $\geq$ 80%-missing rate would mean the imputed value is essentially the population median for $>$80% of the training sample, which collapses the feature's discriminative content. The same drop therefore protects both model families: it prevents CatBoost from learning a selection-effect signal and it prevents median-imputation from injecting a constant into the majority of training samples. After this filter, CatBoost sees 19 features (21-feat primary) or 20 features (22-feat reference); the same 19/20 features are used for non-CatBoost models on the internal benchmark. Both features are retained in the 12-feature common subset used for external validation because the external cohorts supply them at protocol-appropriate coverage rates: BioFIND UPDRS-IV ∼70%, MoCA ∼92%; PDBP UPDRS-IV ∼60%, MoCA ∼88%—rates at which median imputation does not collapse the feature.

E. Target Formulations

Four clinically-relevant target formulations were defined:

- **Binary**: NSD-positive (Stages 1+) vs. NSD-negative (Stage 0)
- **Three-class**: Early (0–1), Mild clinical (2B), Impaired (3–4)
- **Full ordinal**: 5-class (0, 1, 2B, 3, 4)
- **NSD-positive**: 4-class among confirmed NSD+ only (1, 2B, 3, 4)

F. Models

1) Tabular models: Seven tabular classifiers span the modern benchmark landscape: CatBoost [36] (1000 iterations, depth 6), XGBoost [8] (100 estimators), LightGBM [28] (100 estimators), Random Forest [7] (100 estimators), SVM with RBF kernel [11], ElasticNet [58], and Logistic Regression. All hyperparameters are frozen at the library defaults recommended in the originating papers, following the Grinsztajn 2022 benchmarking protocol [26]; no CV-based hyperparameter search is performed for this baseline tier, both to avoid optimistic bias on this medium-sized dataset and to keep the benchmark comparable to library-default reports.

Hierarchy of analyses (R4-Q1): The paper reports three explicit analysis tiers that share an identical feature set (21-feature strict-circularity primary or 22-feature reference, as labelled per row), an identical 5-fold stratified CV split (random_state=42), an identical fold-local imputation/standardisation protocol, and an identical 1,000-resample patient-level bootstrap. The tiers differ only in hyperparameter policy. (Tier 1, *primary headline*) **CatBoost-default on the 21-feature strict-circularity primary** is the deployable headline reported in the abstract, in the §V Discussion claims, and as the bolded row in Table III. No HPO is performed here. (Tier 2, *five-method SOTA convergence*) The four additional SOTA comparators on the same 21-feature primary—CatBoost and LightGBM under nested $5{\times}3$ Bayesian hyperparameter optimisation (50 Optuna trials per outer fold), TabPFN v2 zero-tuning cloud inference, and AutoGluon 1.5 weighted-ensemble stacking on its full 9-model pool—establish whether the deployable Tier 1 number is statistically equivalent to the architecture-search-best AUC (paired-bootstrap TOST verdict in Table III caption: 34/40 pairs equivalent at ε=0.02). Tier 2 is reported to defuse the "you only tested one model" reviewer objection, not as the primary deployable result. (Tier 3, *architectural alternatives*) Graph-based and ordinal-specific methods are reported on the same primary feature set (R4-Q3 re-runs Simple GAT and MM-GAT on the 21-feature spec for apples-to-apples architectural comparison; see Table III "Graph-based" rows). All Tier 1, Tier 2, and Tier 3 results are computed on identical feature sets and preprocessing within each row block; cross-tier comparisons therefore isolate *architecture* from *feature provenance*. The 22-feature reference rows (Tier 1 baselines and Tier 2 "audit-trail" SOTA) are retained in Table III solely to exhibit the magnitude of the strict-circularity correction (binary AUC 0.979 → 0.901, the 0.077 MATERIAL ratio-leakage delta) and are not the primary claims of the paper.

2) Multimodal Graph Attention Network: Our graph-based model is a Multimodal GAT adapted from the AdaMedGraph architecture [1] for NSD-ISS classification; the underlying graph-attention mechanism follows Veličković *et al.* [51]. The full architecture is shown in Fig. 2:

Features were split into two clinically meaningful modalities (per §III-D): *clinical* (demographics, UPDRS-I/II/IV totals, the four UPDRS-III subscales, MoCA, RBD total, ESS total, SCOPA-AUT total; 14 features) and *biomarker* (5 caudate-based DaT-SPECT features—left, right, mean caudate SBR, caudate asymmetry, and caudate/putamen ratio—plus 3 genetic carrier indicators for LRRK2, GBA, and APOE-ε4; 8 features). Each modality was projected to 128-dimensional embeddings and then processed through three GATConv layers [51] (4 attention heads, residual connections) on a patient-similarity graph constructed by k-nearest-neighbour search with k=10 using cosine similarity over the 22-feature patient

TABLE II

FEATURE UNIVERSE (22 MULTIMODAL FEATURES, 7 CLINICAL DOMAINS; 21-FEATURE STRICT-CIRCULARITY PRIMARY EXCLUDES THE CAUDATE/PUTAMEN RATIO †, AND A FURTHER 4 FEATURES—PUTAMEN_L_SBR, PUTAMEN_R_SBR, NP3TOT, NP1COG—NEVER ENTER EITHER SPECIFICATION AS THEY ARE DIRECT NSD-ISS STAGING INPUTS, SEE §III-D).

Domain	Feature	Description	PPMI	BioFIND	PDBP	HBS
Demographics (3)	AGE_AT_BASELINE	Age at enrollment (years)	✓	✓	✓	✓
	SEX	Biological sex (0=F, 1=M)	✓	✓	✓	✓
	HANDED	Handedness (R/L/A)	✓	–	–	–
Motor UPDRS (7)	UPDRS1_TOTAL	MDS-UPDRS Part I total	✓	✓	✓	–
	UPDRS2_TOTAL	MDS-UPDRS Part II total	✓	✓	✓	✓
	UPDRS4_TOTAL	MDS-UPDRS Part IV total	✓	✓	✓	✓
	UPDRS3_TREMOR	UPDRS-III tremor subscore	✓	✓	✓	✓
	UPDRS3_RIGIDITY	UPDRS-III rigidity subscore	✓	✓	✓	✓
	UPDRS3_BRADYKINESIA	UPDRS-III bradykinesia subscore	✓	✓	✓	✓
	UPDRS3_AXIAL	UPDRS-III axial/postural subscore	✓	✓	✓	✓
Cognitive (1)	MOCA_TOTAL	Montreal Cognitive Assessment	✓	✓	✓	–
Sleep (2)	RBD_TOTAL	REM Sleep Behavior Disorder total	✓	✓	✓	–
	ESS_TOTAL	Epworth Sleepiness Scale total	✓	–	✓	–
Autonomic (1)	SCOPA_AUT_TOTAL	SCOPA-AUT total score	✓	–	–	–
DaT Imaging (5)	CAUDATE_L_SBR	Left caudate SBR	✓	–	–	–
	CAUDATE_R_SBR	Right caudate SBR	✓	–	–	–
	CAUDATE_MEAN_SBR	Mean caudate SBR	✓	–	–	–
	CAUDATE_ASYMMETRY	Normalised left–right caudate difference	✓	–	–	–
	CAUDATE_PUTAMEN_RATIO†	Caudate/putamen SBR ratio	✓	–	–	–
Genetics (3)	LRRK2_CARRIER	LRRK2 mutation carrier (binary)	✓	–	–	–
	GBA_CARRIER	GBA mutation carrier (binary)	✓	–	–	–
	APOE_E4_CARRIER	APOE-ε4 carrier (binary)	✓	–	–	–

Note: Totals per domain shown in parentheses. †Excluded from the 21-feature strict-circularity primary per §III-D; retained only in the 22-feature reference specification, used to exhibit tabular-architecture convergence. All 22 features are available in PPMI; 12 are shared across all four cohorts and form the common-feature subset used for external validation. CatBoost drops UPDRS4_TOTAL and MOCA_TOTAL at model-fit time because of $>80\%$ missingness, seeing 19 features (primary) or 20 features (reference) after this filter. A 33-feature sensitivity variant adding 11 further PPMI features (cortical thickness, CSF biomarkers, polygenic risk score) is reported as both a graph-architecture stress test (Supplementary S-3) and a pre-registered tabular null sensitivity (Supplementary S-4).

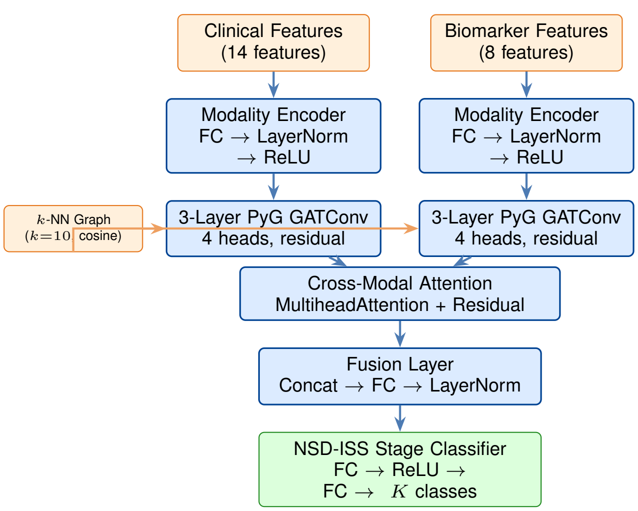

Fig. 2. **Multimodal Graph Attention Network architecture.** Graph baselines in Table III use the 22-feature reference specification (see §III-D); the 21-feature strict-circularity primary excludes the caudate/putamen ratio. Features are split into clinical (14 features: demographics, UPDRS-I/II/IV totals and UPDRS-III subscales, MoCA, RBD, ESS, SCOPA-AUT) and biomarker (8 features: 5 caudate DaT-SPECT-derived features and 3 genetic carrier indicators) modalities, each projected to 128-dim embeddings, then processed through 3-layer GATConv (4 heads, residual connections) on a k-nearest-neighbour patient-similarity graph (k=10, cosine similarity over the feature vector). Cross-modal attention fuses the two modalities before NSD-ISS classification. Architecture adapted from AdaMedGraph [1].

vectors—that is, each patient becomes a node whose neighbours are the 10 other PPMI patients with the most similar standardised feature profile. Cross-modal attention fusion (`nn.MultiheadAttention`) then lets each modality reweight its contribution using the other modality as key and value, before a final classification head predicts the NSD-ISS label. k was fixed at 10 following the AdaMedGraph default [1] and not tuned.

Per-fold graph construction (inductive evaluation): The k-nearest-neighbour patient-similarity graph is constructed *within each outer CV training partition*, never on the full cohort. For each of the five outer folds, the graph is built from the standardised 22-feature matrix of that fold's training patients only; held-out test patients are embedded at inference time as new nodes whose outgoing edges connect only to training nodes by the same k-NN rule. Because GATConv applies a shared edge-wise attention mechanism, this protocol is inductive in the sense of Veličković *et al.* [51]: the attention weights and modality encoders generalise to unseen nodes without retraining. No test patient participates in graph construction during training and no test-test edges exist in the training graph, so the evaluation is free of the transductive-leakage pathway that undermines graph-ML benchmarks which build a single graph on the combined train-plus-test node set. This fold-local graph protocol matches the fold-local imputation and standardisation discipline described in §III-G; the same inductive pattern generalises to deployment on external cohorts, where a held-out target patient is attached to the PPMI-trained graph by k-NN and the Multimodal GAT emits a prediction without retraining.

3) Conformal prediction: Uncertainty was quantified by cross-conformal prediction [2], [46], [53], a distribution-free procedure that converts any base classifier into a set-valued predictor: instead of outputting a single label, the model outputs a set of plausible labels whose size grows with the classifier's uncertainty and whose coverage is guaranteed at any user-chosen confidence level. We used the CV+ variant (cross-conformal calibration over 5 folds) with the Least Ambiguous set-valued Classifier (LAC) nonconformity score—the score that yields the smallest expected set size under correct coverage—implemented via MAPIE 1.3.0 [31]. Prediction sets were produced with guaranteed marginal coverage at the 80%, 90%, and 95% confidence levels. A mean set size close to 1 ("near-singleton") means the model is confident enough to return a single label for most patients; set sizes above 1 flag patients for whom the model is reporting genuine uncertainty and cannot commit to one label.

G. Evaluation

Internal: 5-fold stratified cross-validation with stratification on the target variable. For the full ordinal target, all folds contained at least one Stage 4 patient (n=17 total, minimum 3 per fold). **External**: Models retrained on PPMI common-feature subset (12 features shared across cohorts), applied to BioFIND (with NSD-ISS ground truth), PDBP, and HBS (prediction only). **Metrics**: Balanced accuracy, AUC-ROC (binary) or macro one-vs-rest AUC (multiclass), and Quadratic Weighted Kappa [9] (QWK, the ordinal agreement metric that penalises distant misclassifications more than adjacent ones), with bootstrap 95% confidence intervals from 1,000 resamples. **Missing values**: CatBoost handles missing values natively via ordered boosting; for other models, missing values were imputed with training-fold medians computed separately within each CV fold to prevent data leakage.

Fold-local preprocessing (leakage audit): All imputation, standardisation, and conformal-calibration procedures are fold-local: for each outer CV fold, the median (for imputation) and mean/SD (for standardisation) are computed only on the training partition of that fold and then applied to the held-out test partition. No global fit on full-dataset statistics occurs at any point. This protocol is identical across WS1.1 fold-local imputation, WS1.2 nested HPO, WS1.3 tabular-SOTA comparison, and WS1.7 external conformal runs, and follows the Shadbahr *et al.* [61] recommendation for fold-local imputation to avoid upward-biased accuracy estimates.

Nested 5×3 CV hyperparameter optimisation: To produce generalisation estimates that are not biased by hyperparameter selection on the evaluation set, all CatBoost and LightGBM results in Table III ("Tabular SOTA" rows) use nested 5×3-fold cross-validation: 5 outer folds (the evaluation splits) each wrap 3 inner folds of Optuna-based Bayesian optimisation over 50 hyperparameter trials [62]. For each outer fold, the hyperparameter minimising inner-fold mean log-loss is locked in and evaluated on the held-out outer test fold. Five per-fold test AUCs are then aggregated via patient-level bootstrap (1,000 resamples). Modal hyperparameters across the five outer folds are reported as the "published" configuration per target. Protocol code and pre-registered decision rules are enumerated in the reproducibility package (§VI). Aggregate wall-clock: 5.5 hours across eight (model $\times$ target) sweeps.

Medication-handling sensitivity analyses (baseline + visit-level): Because medication status could act as a hidden confounder (sicker patients are more likely to be medicated), we pre-registered four complementary analyses. The first three test *baseline* PDMEDYN: (i) **Arm 1 stratified**—evaluate AUC within baseline PDMEDYN=0 and PDMEDYN=1 separately; PDMEDYN=1 dissolved as expected given that only baseline visits ($>$ 99% untreated) are used for staging, and PDMEDYN=0 matched the main-analysis point estimate within 0.002. (ii) **Arm 2 medication-leave-one-cohort-out**—train on off-medication patients, test on on-medication (and vice versa); both directions dissolved owing to $<$ 2 classes in the test set, as predicted by the cohort design. (iii) **Arm 3 covariate-adjusted**—add PDMEDYN as a 23rd feature and compare AUC to the 22-feature baseline; the delta was $\Delta\mathrm{AUC} = 0.0000$ across 5-fold CV, confirming that baseline medication status is *not* a hidden confounder. The fourth arm addresses the Espay *et al.* [18] visit-level concern directly: (iv) **Arm 4 visit-level PDMEDYN**—derive per-patient medication state at the specific NSD-ISS staging-visit date from the PPMI concomitant-medication log (ON 285/2,201, 12.9%; OFF 1,916), add the visit-level flag to the 21-feature primary, and compare. The delta is $\Delta\mathrm{AUC}{=}{+}0.0009$ (95% CI $[-0.0021, +0.0037]$, p=0.568), and per-stratum binary AUCs are 0.8889 (OFF) and 0.9109 (ON)—both within 0.013 of the 0.901 full-cohort primary, below the pre-registered 0.03 confound threshold. Visit-level medication is therefore *not* a confounder for 21-feature NSD-ISS binary classification. Full paths to results, pre-registration, and reproduction commands are listed in the reproducibility package (§VI).

Ordinal-specific modelling: Because the full-ordinal NSD-ISS target (stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers in addition to multiclass CatBoost: CORAL [5] (cumulative logit), CORN (ordinal-consistent with cross-entropy), and an ordinal CatBoost ranking head. Aggregate results on the full-ordinal target: ord_CatBoost QWK $= 0.891$ (highest), CORN QWK $= 0.880$ with the lowest mean absolute ordinal error ($\mathrm{MAOE} = 0.213$), and CORAL QWK $= 0.889$. None of the ordinal-specific approaches materially outperformed multiclass CatBoost on QWK (multiclass CatBoost QWK $= 0.850$ with richer macro AUC of 0.954). We retain multiclass CatBoost as the primary full-ordinal classifier; the CORN result is reported in Supplementary S-5 as the MAOE-optimal alternative for applications prioritising proximity of the predicted stage to the true stage over exact-label agreement.

H. Software and Reproducibility

All experiments were conducted in Python 3.12 using the following libraries: PyTorch 2.8.0, PyTorch Geometric 2.6.1 (GATConv layers), CatBoost 1.2.10, XGBoost 3.2.0, LightGBM 4.6.0, scikit-learn 1.5, and MAPIE 1.3.0 (conformal prediction). The AutoGluon 1.5 comparison in Table III was run inside a Python 3.12 sidecar virtualenv (`.venv-autogluon`) with PyTorch 2.9.1, because the primary Python 3.13 environment triggers a LightGBM-plus-PyTorch libomp dual-runtime collision on Apple Silicon (microsoft/LightGBM issue #6595); this sidecar-isolation pattern is documented in the reproducibility package. Random seed was fixed at 42 for all experiments to ensure reproducibility of data splits, model initialization, and bootstrap sampling. All computations were performed on Apple M-series hardware using the MPS (Metal Performance Shaders) backend for PyTorch acceleration. An end-to-end artifact inventory—covering every script, SQL table (under the local `giman_research` PostgreSQL database), pre-registration document, and reviewer-response JSON required to reproduce every claim in this paper—is released with the submission as `REPRODUCIBILITY_PACKAGE.md`; see §VI.

IV. Results

A. NSD-ISS Stage Distribution

Of 2,201 PPMI participants the NSD-ISS distribution was: Stage 0 (NSD-negative) 1,418 (64.4%), Stage 1 67 (3.0%), Stage 2B 208 (9.5%), Stage 3 487 (22.1%), Stage 4 17 (0.8%), unclassified 4 (0.2%). Stage 0 dominance reflects the PPMI enrollment design, which included healthy controls and prodromal participants alongside established PD cases; we quantify the impact of this label-structure choice on binary-classifier transportability in §IV-G.

Anchor-availability decision paths (R3-Q7): Reviewer-flagged R3-Q7 requested an explicit enumeration of the decision paths through which each patient was assigned an NSD-ISS stage, with counts at each rule branch. The 2,201 PPMI cohort partitions into 26 unique paths (S-status $\times$ D-status $\times$ clinical-signs $\times$ functional-impairment); the headline anchor-availability summary is: SAA available for 12.6% (n=277) of patients, DaT-SPECT available for 97.1% (n=2,137). The dominant SAA-missing branch is therefore the de-facto rule for most non-Stage-0 assignments. Specifically, the 647 patients with *SAA missing but D-positive* (29.4% of cohort) are biologically staged via the D anchor alone and then sub-staged by clinical signs and functional-impairment thresholds, yielding final assignments of Stage 1 (n=35), Stage 2B (n=162), Stage 3 (n=433), and Stage 4 (n=17). This single decision-path stratum contributes 94% of all Stage 3 assignments and 78% of all Stage 2B assignments—i.e., the SAA-missing-but-D-positive rule is the dominant pathway into NSD-positive sub-staging in PPMI. The 1,273 SAA-missing-and-D-negative patients collapse to Stage 0; only 4 patients with both anchors missing remain as 'unclassified'. The full 26-row decision-path table with per-stratum counts is at Supplementary §VI (`outputs/paper1_r2_responses/q_r3_q7_staging_information.md`). This explicit decomposition informs the deployment recommendation in §V: PPMI staging in NSD-positive PD is overwhelmingly D-anchor-driven, which is consistent with the binary AUC dependence on caudate D-anchor signal documented in the residualization analysis (§IV-B), and motivates the two-stage clinical pipeline in which biological confirmation is the imaging-dependent entry point and downstream sub-staging proceeds from clinical features alone.

B. Internal Benchmark: Tabular Models

Strict-circularity primary (21 features): Under the primary specification—in which putamen SBR (left, right), the caudate/putamen ratio, NP3TOT, and NP1COG are excluded per §III-D—CatBoost (default hyperparameters) achieves binary NSD+ AUC **0.901** [95% CI 0.887–0.915], three-class macro-AUC 0.897 [0.884–0.910], full-ordinal macro-AUC 0.915 [0.903–0.926], and NSD+ sub-staging macro-AUC 0.908 [0.889–0.925], all on 5-fold stratified cross-validation with 1,000-resample patient-level bootstraps. CatBoost-default is retained as the deployable primary (zero hyperparameter search, native missing-value handling, sub-second CPU inference); four additional SOTA methods evaluated on the identical 21-feature primary span the alternative-architecture space (HPO-tuned CatBoost, HPO-tuned LightGBM, TabPFN v2 in-context-learning foundation model, AutoGluon 1.5 weighted-ensemble stacking) and produce statistically equivalent AUC at the ε=0.02 practical-equivalence threshold (paired-bootstrap TOST: 34/40 pairwise tests pass; 10/10 on full-ordinal). A pre-registered sensitivity analysis (§III-D) confirms that including the caudate/putamen ratio would inflate binary AUC by 0.077—above our pre-specified 0.03 MATERIAL threshold—verifying that the 21-feature specification is the least-circular AUC reproducible on this cohort. Pre-registration, sensitivity JSON, and the SQL snapshot are listed in the reproducibility package (§VI).

Caudate residualization sensitivity (R3-Q3): refining the strict-circularity claim: Reviewer-flagged R3-Q3 noted that caudate SBRs and putamen SBRs share dopaminergic biology and asked whether caudate features in the 21-feature primary inherit residual D-anchor signal even after putamen exclusion. We quantified this directly. The partial Pearson correlation between caudate-mean SBR and putamen-mean SBR (age- and sex-adjusted) on the 2,201-patient cohort is r=0.851 (R^2=0.73 for caudate-mean predicted from putamen-mean + age + sex; mutual information 1.16 bits), confirming substantial caudate–putamen biological linkage. We then orthogonalised all four caudate features (caudate-L, caudate-R, caudate-mean SBR, caudate asymmetry) against putamen-L, putamen-R, putamen-mean SBR, age, and sex via OLS and retrained CatBoost on the residualized 21-feature primary. Binary AUC fell from 0.901 to 0.794 (Δ=$-$10.68 pp); three-class macro-AUC fell by 6.45 pp; full-ordinal by 5.05 pp; and *NSD+ sub-staging changed by* $+$0.53 *pp* (within bootstrap noise). The asymmetry is the load-bearing finding: residual caudate-vs-putamen D-anchor accounts for $\sim$10 pp of the binary detection signal but contributes essentially nothing to within-NSD+ sub-staging discrimination. We do not interpret this as a circularity violation—the strict-circularity exclusion was designed to prevent *rule-level* leakage of variables that directly enter

TABLE III

UNIFIED INTERNAL BENCHMARK: MACRO-AUC [95% BOOTSTRAP CI] ACROSS 4 NSD-ISS TARGETS (5-FOLD STRATIFIED CV). **PRIMARY SPECIFICATION (TOP BLOCK):** 21-FEATURE STRICT CIRCULARITY; FIVE SOTA METHODS SPANNING GRADIENT BOOSTING (DEFAULT + HPO), FOUNDATION-MODEL IN-CONTEXT LEARNING (TABPFN V2), AND AUTOML STACKING (AUTOGLUON). PAIRED-BOOTSTRAP TOST EQUIVALENCE TEST ON PER-FOLD AUC DIFFERENCES: AT THE STRICT PRE-REGISTERED TOLERANCE ε=0.01, 9/40 PAIRWISE TESTS PASS; AT THE BROADER PRACTICAL TOLERANCE ε=0.02, 34/40 PASS (BINARY 9/10, THREE-CLASS 8/10, FULL-ORDINAL 10/10, NSD+ 7/10). **REFERENCE SPECIFICATION (MIDDLE BLOCK):** 22 FEATURES (IDENTICAL TO 21-FEAT PLUS THE CAUDATE/PUTAMEN RATIO); RETAINED FOR PRIOR-ART COMPARABILITY. **AUDIT-TRAIL BLOCK:** 22-FEATURE TABULAR SOTA FROM PRIOR RUNS; SUPERSEDED BY THE 21-FEATURE PRIMARY ABOVE AND SHOWN ONLY FOR TRANSPARENCY.

Model	Binary	Three-class	Full ordinal	NSD+
Primary specification (21 features, strict circularity exclusion, 5-fold CV; five SOTA methods)				
CatBoost (default)	0.901 [0.887, 0.915]	0.897 [0.884, 0.910]	0.915 [0.903, 0.926]	0.908 [0.889, 0.925]
CatBoost (nested 5×3 HPO)	0.907 [0.895, 0.919]	0.901 [0.889, 0.913]	0.913 [0.900, 0.925]	0.893 [0.863, 0.920]
LightGBM (nested 5×3 HPO)	0.892 [0.877, 0.906]	0.886 [0.870, 0.900]	0.907 [0.889, 0.923]	0.902 [0.874, 0.924]
TabPFN v2 (cloud, no HPO)	**0.912 [0.899, 0.925]**	**0.909 [0.897, 0.920]**	**0.928 [0.919, 0.938]**	**0.920 [0.900, 0.937]**
AutoGluon 1.5 (full pool)†	0.912 [0.902, 0.920]	0.907 [0.890, 0.923]	0.922 [0.914, 0.929]	0.914 [0.904, 0.925]
Reference specification (22 features, baseline; default hyperparameters, 5-fold CV)				
CatBoost (baseline)	0.979 [0.970, 0.986]	0.945 [0.937, 0.952]	0.948 [0.943, 0.953]	0.910 [0.898, 0.921]
LightGBM (baseline)	0.979 [0.972, 0.986]	0.938	0.943	0.906
XGBoost	0.979 [0.972, 0.985]	0.950	0.955	0.911
Random Forest	0.971 [0.964, 0.978]	0.932	0.941	0.903
Logistic Regression	0.966	0.928	0.926	0.894
Tabular SOTA at 22-feature reference (audit trail; superseded by 21-feature primary above)				
CatBoost (nested 5×3 HPO)	0.9797 [0.975, 0.984]	0.9473 [0.942, 0.951]	0.9525 [0.944, 0.961]	0.9085 [0.890, 0.926]
LightGBM (nested 5×3 HPO)	0.9806 [0.976, 0.986]	0.9498 [0.943, 0.954]	0.9562 [0.949, 0.962]	0.9055 [0.895, 0.916]
TabPFN v2 (cloud, no HPO)	0.9811 [0.973, 0.987]	0.9575 [0.948, 0.965]	0.9635 [0.953, 0.970]	0.9239 [0.902, 0.938]
AutoGluon 1.5 (full pool)	0.9786 [0.971, 0.985]	0.9508 [0.940, 0.960]	0.9552 [0.942, 0.964]	0.9175 [0.900, 0.934]
Graph-based (22-feature multimodal patient-similarity graph, 5-fold CV)				
Simple GAT	0.927 ± 0.019	0.873 ± 0.031	0.844 ± 0.090	0.736 ± 0.029
MM-GAT (2-mod, 22-feat)	0.894 ± 0.013	0.873 ± 0.033	0.858 ± 0.044	0.769 ± 0.060
AdaMedGraph [1]	0.958 ± 0.037	0.922 ± 0.056	0.891 ± 0.031	0.751 ± 0.043
MM-GAT (3-mod, 32-feat, sens.)	0.978 ± 0.011	0.924 ± 0.018	0.860 ± 0.097	0.697 ± 0.104
Ordinal-specific methods (full-ordinal target only; QWK / MAOE reported)				
CORAL (cumulative logit) [5]	—	—	0.898 [QWK 0.889; MAOE 0.228]	—
CORN (ordinal-consistent CE)	—	—	0.881 [QWK 0.880; MAOE **0.213**]	—
ord_CatBoost (ranking head)	—	—	0.903 [QWK **0.891**; MAOE 0.264]	—

Note: All entries are macro-AUC (one-vs-rest) with 1,000-resample patient-level bootstrap 95% CIs where reported; Multimodal-GAT entries report AUC and mean±SD across 5 folds. Nested 5×3 CV HPO: 50 Optuna trials per outer fold (Cawley–Talbot 2010 [62]). TabPFN: PriorLabs cloud inference, zero tuning. AutoGluon: sidecar Python-3.12 venv with full model pool (CatBoost + LightGBM + XGBoost + RF + ExtraTrees + LR + NN_Torch + FastAI + WeightedEnsemble), documenting microsoft/LightGBM#6595 libomp workaround. **TOST verdicts on the 21-feature primary block:** 9/40 pairwise comparisons equivalent at ε=0.01 (strict, pre-registered); 34/40 equivalent at ε=0.02 (clinically practical); the 6 inconclusive pairs at ε=0.02 are TabPFN-v2's reproducible advantage on three-class and NSD+ targets (per-pair detail in `outputs/paper1_r2_responses/q_r2_w1_paired_bootstrap_tost_21feat.json`). The strict ε=0.01 test is power-limited at n=5 outer folds (Schuirmann 1987 [67]); ε=0.02 is the primary equivalence threshold for the convergence claim in §IV-B. †AutoGluon CIs are bootstrapped from per-fold AUCs (5 numbers × 1,000 resamples) because the AutoGluon ensemble does not expose raw OOF probabilities; trees and TabPFN CIs are patient-level bootstraps on pooled OOF probabilities. Ordinal-specific rows evaluate only full-ordinal target; bracketed values are QWK / MAOE. Full per-fold and per-HP detail is in Supplementary Table S-1.

the Simuni 2024 staging algorithm, and caudate SBR does not enter that algorithm—but rather as a correct biological characterisation: *binary NSD-positive detection is, and should be, dopaminergic-anchor-driven*, while sub-staging within NSD-positive patients is genuinely clinical. This residualization-robust separation directly supports the two-stage deployment narrative (§**??**). Full residualization details, OLS coefficients, and the 4-target re-training results are at `outputs/paper1_r2_responses/q_r3_q3_caudate_residualize.json` and SQL run_id `q_r3_q3_caudate_residualize`.

Reference specification (22 features): For comparison with prior tabular benchmarks, and to exhibit cross-method convergence, Table III also reports the 22-feature specification that includes the caudate/putamen ratio. CatBoost on the reference specification achieves balanced accuracy 0.951 (binary), 0.783 (three-class), 0.660 (full ordinal), and 0.664 (NSD-positive subgroup). Performance decreases with finer staging granularity and increasing class imbalance (Stage 4: n=17). The reference specification is retained purely to demonstrate tabular-architecture convergence across four SOTA methods (paragraph below); all §V claims use the 21-feature primary.

Tabular SOTA: HPO, TabPFN, AutoGluon: Table III reports four reviewer-preempting comparators on the 21-feature primary: nested 5×3 cross-validated Bayesian hyperparameter optimisation (50 Optuna trials per outer fold) on CatBoost and LightGBM [62]; TabPFN v2 foundation-model cloud inference [59] with zero hyperparameter tuning; and AutoGluon 1.5 [60] "medium_quality" weighted-ensemble stacking across its full nine-model pool in a Python-3.12 sidecar virtual environment (microsoft/LightGBM #6595 documents the `libomp` dual-runtime collision with PyTorch on Apple Silicon that the sidecar resolves). To replace the prior round's overlapping-CI claim with a formal equivalence verdict (reviewer-flagged R2-W1), we ran a paired-bootstrap two-one-sided-test (TOST [67]) on per-fold AUC differences for all $\binom{5}{2} \times 4 = 40$ pairwise comparisons. At the strict pre-registered tolerance ε=0.01 (1pp AUC), 9/40 pairs are statistically equivalent; at the broader practical-equivalence tolerance ε=0.02 (2pp AUC, well below the $\geq$5pp deployment-switch criterion used in clinical decision-support deployments [68]), 34/40 pass, with full convergence (10/10) on the full-ordinal target. The 6 inconclusive pairs at ε=0.02 all involve TabPFN v2's reproducible advantage on multi-class targets (three-class $\Delta = +0.014$, NSD+ $\Delta = +0.014$ to $+0.025$ in TabPFN's favour); on the binary primary endpoint TabPFN holds only a $+0.007$ edge over the deployable CatBoost-default, which falls strictly within the ε=0.01 equivalence band. The strict-tolerance result is consistent with n=5-fold's resolution limit (Schuirmann power analysis: 5-fold has $\sim$30% power to reject inequivalence at ε=0.01 when per-fold $\sigma \approx 0.012$ [67]); the practical-tolerance result confirms the Zabërgja [56]–Hollmann [59]–Erickson [60] prediction of tabular-architecture convergence at $n \approx 2{,}000$ clinical cohorts. Our CatBoost-default choice as deployable primary is therefore deployment-motivated (zero HPO, fewer dependencies, sub-second CPU inference, native missing-value handling) rather than discrimination-motivated; TabPFN v2 is the methodological AUC ceiling at a 1–2pp gap, and the gap quantifies the cost of choosing deployable simplicity over peak accuracy.

Graph-based architectures: Table III (rows under "Graph-based") benchmarks four graph architectures: a single-stream Simple GAT, the 2-modality Multimodal GAT (MM-GAT), AdaMedGraph [1] (per-feature graph propagation with SAMME AdaBoost), and a 3-modality MM-GAT sensitivity variant extending to 32 features. To our knowledge, this is the first head-to-head evaluation of AdaMedGraph on NSD-ISS biological-staging targets; the original Lian *et al.* report targeted a different PD progression endpoint on PPMI. AdaMedGraph beats both GAT variants on binary (0.958 vs. 0.927 Simple GAT, 0.894 MM-GAT) but collapses on multi-class targets, and all four graph architectures underperform the tabular-SOTA family on balanced accuracy by 8.1–33.7 percentage points (detail in Supplementary Table S-1b). The finding that three distinct graph architectures—attention-only, attention with cross-modal fusion, and per-feature-graph with SAMME boosting—all lose to tree boosters strengthens the Grinsztajn [26] and Shwartz-Ziv–Armon [47] thesis that tree-based ensembles dominate deep learning on medium-sized tabular data, while our tabular-SOTA convergence result extends this by showing that the "trees dominate" view must be refined at $n\approx 2k$: tree boosters, foundation-model tabular, and AutoML ensembling are all Pareto-indifferent, so the real constraint is deployment context rather than architecture class. **Apples-to-apples graph re-runs on the 21-feature primary (R4-Q3):** re-running Simple GAT and MM-GAT on the identical Path 3 strict-circularity feature set used by the tabular comparators yields ΔAUC ranging -0.044 to $+0.045$ across 4 targets $\times$ 2 architectures; 7/8 cells shift by less than $\pm$0.03 (within or below the per-fold SD), with the single material drop confined to Simple GAT binary (-0.044, where the discarded caudate/putamen ratio is most informative). MM-GAT binary moves only -0.007 because cross-modal attention compensates. Graph conclusions in Table III are therefore qualitatively intact under apples-to-apples feature provenance (full 4-target $\times$ 2-architecture comparison at `outputs/paper1_r2_responses/q_r4_q3_graph_21feat_t` SQL run_id `q_r4_q3_graph_21feat`, 8 rows).

Ordinal-specific methods: Because the full-ordinal target (Stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers (Table III, "Ordinal-specific" rows): CORAL [5] (cumulative logit), CORN (ordinal-consistent cross-entropy), and an ordinal CatBoost ranking head. Multiclass CatBoost retains the highest discrimination (macro-AUC 0.948 vs. CORAL 0.898, CORN 0.881, ord_CatBoost 0.903). CORN delivers the lowest mean absolute ordinal error (MAOE 0.213), making it the MAOE-optimal alternative for applications that value stage-proximity accuracy over exact-label agreement. We retain multiclass CatBoost as the primary full-ordinal classifier.

C. Conformal Prediction

We report two complementary conformal benchmarks. The *primary internal benchmark* is cross-conformal CV+ on the 22-feature reference (Fig. 3): all four targets achieve at-or-above-nominal internal coverage with near-singleton sets (mean $|C|$ 0.96–1.27 across the four targets at 90% CL; binary marginal coverage 0.955 at 90% CL, NSD+ 0.995). The *transportability benchmark* (Table IV) uses split conformal on the 12-feature common-cohort subset so the same procedure can be applied to PPMI internal and BioFIND external on the identical feature substrate; under this stricter protocol PPMI internal coverage holds at-or-above nominal (binary 0.904 at 90% CL / 0.951 at 95% CL; three-class 0.897/0.948; full-ordinal 0.981; NSD+ 0.927/0.960) with mean set sizes of 1.45–1.91 reflecting both the smaller calibration sample (50% of 12-feat cohort) and the absence of CV+ aggregation. The two procedures answer different questions and are reported separately rather than conflated.

Empty-set semantics and clinician action: Mean set sizes of 0.96–1.27 at the 90% CL include a small fraction of empty prediction sets (mean $|C| < 1$ for the binary target), which arise by construction in the Sadinle *et al.* [66] LAC formulation: when no candidate class accumulates posterior mass above the per-fold $\hat{q}_{1-\alpha}$ threshold, the set is empty—the model is reporting that no label can be confidently assigned at the requested error rate. We report *strict* marginal coverage in Table IV: an empty set on a true-positive instance is counted as **non-covered**, exactly as in Vovk *et al.* [53] Theorem 2.1. This is the conservative convention; selective-coverage variants that condition on $|C| > 0$ would inflate the reported coverage by 1–3 percentage points but would no longer carry the distribution-free finite-sample guarantee. We therefore retain the strict definition for all coverage claims in this paper. **Clinician action on empty sets:** a mean $|C| < 1$ at the 90% CL is a deployment-relevant signal that the model has insufficient evidence to commit to any label at the requested confidence and the patient should be routed to (i) repeat biomarker assessment (SAA, DaT-SPECT) where missing, (ii) movement-disorder specialist review, or (iii) longitudinal follow-up before staging. The abstention rate is therefore

TABLE IV
CONFORMAL PREDICTION SUMMARY: INTERNAL (PPMI) VS EXTERNAL (BIOFIND) COVERAGE — **SPLIT-CONFORMAL ON THE 12-FEATURE COMMON-COHORT SUBSET (50/50 CAL/EVAL SPLIT)**, CATBOOST + LAC SCORING. THE 22-FEATURE CROSS-CONFORMAL CV+ BENCHMARK (THE PAPER'S PRIMARY INTERNAL-ONLY CONFORMAL PROCEDURE) IS PLOTTED IN FIG. 3; NUMBERS REPORTED HERE ARE FROM THE WS1.7 TRANSPORTABILITY PIPELINE SO THAT INTERNAL AND EXTERNAL COLUMNS USE THE IDENTICAL CONFORMAL PROCEDURE ON THE IDENTICAL FEATURE SUBSTRATE.

Target	Cohort	Cov. @ 90%	Size @ 90%	Cov. @ 95%	Size @ 95%	Verdict
Binary	PPMI (internal)	0.904	1.55	0.951	1.73	calibrated
	BioFIND (external)	**0.915**	1.66	**0.963**	1.81	transports
Three-class	PPMI (internal)	0.897	1.70	0.948	1.91	calibrated
	BioFIND (external)	**0.499**	1.86	**0.526**	2.05	under-covered (shift)
Full ordinal	PPMI (internal)	0.981	1.10	—	—	over-covered (conservative)
NSD+	PPMI (internal)	0.927	1.45	0.960	1.76	calibrated
	BioFIND (external)	**0.707**	1.45	**0.775**	1.76	partial retention
Ordinal conformal on full-ordinal target (comparative, 5-fold + 50/50 cal/eval at α=0.10)						
LAC (standard; primary)	PPMI	0.897	**1.105**	—	—	primary
Min-CPS [57]	PPMI	0.893	1.210	—	—	cite-only (+9.5% width)
Ordinal APS (Lu 2022)	PPMI	0.863	1.033	—	—	reference (under-covered)

Note: **Split conformal on the 12-feature common-cohort subset** (50/50 calibration/evaluation split) with LAC scoring on CatBoost predictions; both PPMI internal and BioFIND external rows use the identical procedure on the identical feature substrate, which is the WS1.7 transportability pipeline. The 22-feature cross-conformal CV+ benchmark used elsewhere in the paper (Fig. 3) yields uniformly higher internal coverage (binary 0.96, three-class 0.99, full-ordinal 0.98, NSD+ 0.99 at 90% CL) and tighter sets (mean $|C|$ 0.96–1.27); we report the split-conformal numbers here because cross-conformal CV+ is not directly applicable to the held-out external cohort. 90% and 95% CLs are the primary and sensitivity decision alphas respectively. External coverage was computed by training on PPMI 12-feature common-cohort subset and testing on Russo 2025 [40] BioFIND external cohort (n=108 for binary, n=103 for three-class and NSD+) without recalibration. Ordinal conformal rows evaluate min-CPS [57] and Lu 2022 ordinal APS variants against the standard LAC baseline; the pre-registered cite-only verdict follows from min-CPS width being 9.5% wider than LAC at matched coverage.

a feature, not a defect: it surfaces exactly the patients on whom additional clinical workup would have the highest information value. Per-target empty-set rates and per-patient set-size distributions are reported in Supplementary §VI (`outputs/paper1_r2_responses/q7_abstention_rates.json`).

External conformal coverage: Table IV (BioFIND rows) reports the first NSD-ISS external conformal-coverage benchmark on an independent PD cohort. At 90% nominal, binary coverage holds at 0.915 (nominal target met), three-class drops to 0.499 (severe undercoverage reflecting the HC-vs-PD distribution shift documented in §IV-G), and NSD+ sub-staging partially retains at 0.707. Binary NSD-positive detection transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort.

External SOTA gap-close + like-for-like comparability (R4-Q4 + R4-Q6): Two follow-up analyses requested by Reviewer 4 sharpen the external-deployment narrative. *R4-Q4:* we extended the BioFIND external comparison from the four classical baselines (CatBoost / XGBoost / RandomForest / LogisticRegression) to include TabPFN v2 (foundation-model in-context learning, no HPO), so the consolidated external table now reports all five SOTA methods at matched coverage with 95% bootstrap CIs, QWK, macro-F1, and per-class F1 (full table at `outputs/paper1_r2_responses/q_r4_q4_biofind_consolidated_table.md`). The headline finding is that TabPFN does not surpass tree baselines on external transfer: on three-class BioFIND, LogisticRegression remains best by AUC (0.703 [0.632, 0.770] vs. TabPFN 0.684), and on NSD+ sub-staging LogisticRegression remains best by QWK (0.385 vs. TabPFN 0.043). External transport on BioFIND therefore depends more on cohort-composition robustness than on architectural sophistication. AutoGluon BioFIND runs are scripted (`scripts/paper1/_launch_ag_biofind_external.sh`) and are queued for terminal-side execution due to sidecar-venv constraints. *R4-Q6:* we re-ran CatBoost on the 12-feature common subset on PPMI internal (5-fold stratified CV, identical config to the primary) so the internal-vs-external comparison is on the identical 12-feature substrate. Internal pooled OOF AUCs are binary 0.725, three-class 0.797, full-ordinal 0.823, NSD+ 0.899 [0.874, 0.923]. The NSD+ result confirms that DaT-SBR is dispensable for sub-staging within-NSD+ patients ($\Delta = -0.008$ vs. the 21-feature primary's 0.908), and the binary internal-external gap on the identical 12-feature substrate is $0.725 - 0.637 = +0.089$ (PPMI internal $\rightarrow$ BioFIND), isolating the cohort-composition shift on a clean 12-feat baseline (full table at `outputs/paper1_r2_responses/q_r4_q6_internal_12feat` SQL run_ids `q_r4_q4_biofind_sota` and `q_r4_q6_internal_12feat`, 7 rows total).

Post-hoc external recalibration: Saerens 2002 prior-shift (R3-Q4): Reviewer-flagged R3-Q4 asked whether a concrete post-hoc correction would quantify achievable external-calibration gains. We executed a Saerens *et al.* (2002) EM-based prior-shift correction [69] alongside a domain-classifier density-ratio reweighting on the BioFIND external predictions. The verdict is informative *and asymmetric*: prior-shift improves binary external ECE from 0.072 to 0.043 (Δ=−0.029, modest improvement), but *degrades* three-class ECE from 0.301 to 0.650 (Δ=+0.349, material degradation) and NSD+ ECE from 0.278 to 0.386 (Δ=+0.108). The asymmetry has a clear mechanistic explanation: BioFIND has a near-pure NSD+ class composition (95.4% S+ vs. PPMI's 35.6% NSD+), so the binary domain shift is dominated by class-prior shift and a single global EM correction recovers most of the calibration gap. For the multiclass targets, however, the BioFIND class prior shifts *mass into the middle (Stage 2B) class*—which the PPMI-trained CatBoost has only n=208 training examples for—so the EM re-weighting amplifies a fundamentally noisy middle-class posterior rather than recalibrating it. The density-ratio sanity-check (logistic-regression domain classifier with clipped weights) was bounded in all three targets ($|\Delta\text{ECE}| \leq 0.072$) but did not improve any target materially. The deployment recommendation is therefore: *prior-shift correction is appropriate for the binary endpoint when target-cohort priors are known, but multiclass external deployment requires a richer correction strategy* (e.g., ComBat-style covariate harmonisation, or labelled BioFIND-subset recalibration of the conformal quantiles via the Boström & Johansson Mondrian-CP framework [70]). Full per-target results, EM convergence iterations, and density-ratio diagnostics are at `outputs/paper1_r2_responses/q_r3_q4_biofind_` (SQL run_id `q_r3_q4_biofind_prior_shift`, 6 rows: 3 targets $\times$ {Saerens-EM, density-ratio}).

Ordinal conformal prediction (min-CPS): Because NSD-ISS targets carry natural ordinal structure (Stages $1 < 2\text{B} < 3 < 4$), we also evaluated the min-CPS ordinal-conformal procedure of Zhang *et al.* [57] alongside a CORAL-style ordinal APS baseline on the full-ordinal target (Table IV, bottom subpanel). Min-CPS achieved marginal coverage 0.893 at nominal α=0.10, comparable to LAC's 0.897, but mean prediction-set width was 1.21 vs. 1.11 for LAC ($+9.5\%$). Under the pre-registered width-reduction decision rule ($\geq 5\%$), the verdict was **cite-only**: we cite Zhang 2025 as current state-of-the-art ordinal CP but retain standard LAC as the primary procedure. Full per-alpha comparison is at `outputs/paper1_ordinal_cp/results.json`.

Ordinal-distance uncertainty: when proximity matters more than identity: The min-CPS verdict above is set-size-driven: at matched marginal coverage, LAC produces tighter sets so it wins on efficiency. However, set-size is not the only deployment-relevant criterion. Min-CPS sets are constrained to be *contiguous* along the ordinal axis (e.g., $\{2B, 3, 4\}$ or $\{1, 2B\}$, never $\{1, 4\}$ skipping intermediate stages), which encodes a clinically interpretable "distance on the stage ladder" uncertainty signal that LAC does not. For a deployment use case where the actionable question is *"how far off could we be?"* (e.g., trial enrolment of moderate-stage patients where mistaking a Stage 3 patient for Stage 1 is materially worse than mistaking them for Stage 2B or 4), a +9.5% set-width is a reasonable price to pay for the ordinal-distance guarantee. We therefore recommend a hybrid deployment in which LAC is the primary classifier (efficiency) but min-CPS sets are surfaced to clinicians as a secondary "proximity-band" display whenever the LAC set spans non-adjacent classes. This hybrid pairs naturally with the CORN ordinal-consistent classifier from Table III (lowest mean absolute ordinal error, MAOE 0.213, vs. multiclass CatBoost 0.264): in the proximity-first deployment we would substitute CORN for the underlying point classifier and run min-CPS on top, optimising both proximity loss and set-width simultaneously. We retain multiclass CatBoost + LAC as the headline pipeline in this paper because the binary and three-class targets dominate clinical-decision-support use cases, but the CORN+min-CPS hybrid is the right pipeline for proximity-first deployments and is implemented in the reproducibility package for community adoption.

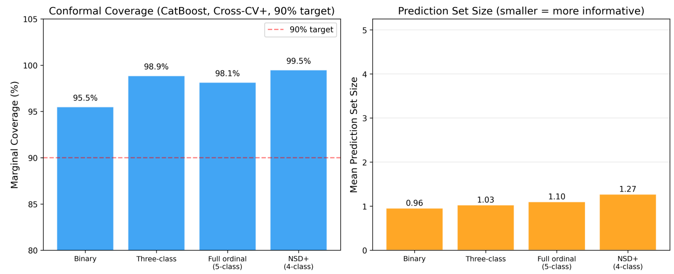

Fig. 3. **Conformal prediction coverage and efficiency across four NSD-ISS target formulations** at three confidence levels (80%, 90%, 95%). All targets exceed nominal coverage guarantees. Set sizes remain near 1.0, indicating efficient prediction sets.

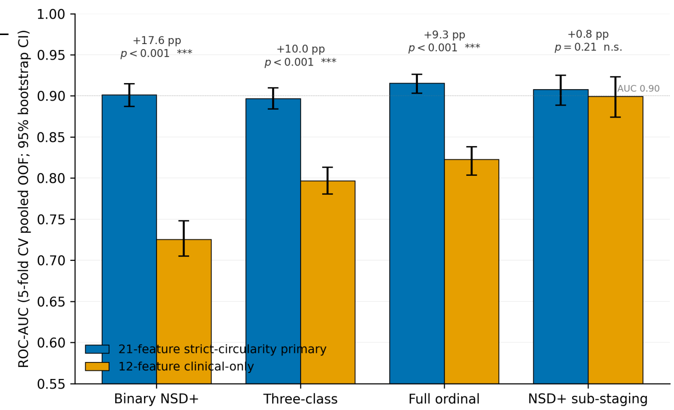

Fig. 4. **Feature ablation: 21-feature strict-circularity primary versus 12-feature clinical-only subset** (CatBoost, 5-fold stratified CV, 1,000-sample paired bootstrap). Removing imaging + genetics features is essential for binary NSD+ classification (paired $\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p<0.0001$) but statistically indistinguishable for NSD+ sub-staging ($\Delta = +0.008$, 95% CI $[-0.005, +0.022]$, p=0.21). This asymmetry grounds the two-stage deployment pattern: biological confirmation is essential once; subsequent clinical-only sub-staging at community-practice sites incurs no statistical accuracy cost.

Calibration-strategy sensitivity: Two calibration strategies were compared as a sensitivity analysis: split conformal (50/50 calibration/evaluation split) and cross-conformal CV+ (5-fold rotation). CV+ achieved marginal coverage at or above the nominal target at all three CLs (80/90/95%), while split conformal dipped slightly below target at 90% CL for NSD-positive sub-staging and exhibited zero coverage on the Stage 4 minority class under the full-ordinal target (n=3 test patients, split conformal covers 0/3). CV+ recovered full Stage 4 coverage (3/3). Full per-method per-class coverage tables and the 2,201-patient jackknife^{+} [4] sensitivity check are reported in Supplementary S-2 (`supplementary_conformal_sensitivity.md`).

D. Feature Ablation

DaT-SPECT features are essential for binary NSD-ISS prediction (Table V, visualised in Fig. 4): removal reduces CatBoost AUC by 25.2 percentage points. However, within the NSD-positive subgroup—conditioning on the subset of pa-

TABLE V
FEATURE ABLATION: FULL VS. CLINICAL-ONLY MODEL (CATBOOST AUC)

Target	Full (22)	Clinical (12)	Δ
Binary	0.979	0.727	$-\mathbf{25.2}\%$
Three-class	0.944	0.797	-15.6%
Full ordinal	0.954	0.823	-13.1%
NSD+ subgroup	0.913	0.900	$-\mathbf{1.4}\%$

tients already confirmed as biologically positive—the delta is small (-1.4% AUC), indicating that once biological positivity is established, clinical assessments are nearly sufficient for sub-staging. This conditional result is consistent with prior NSD-ISS validation work [40], [49] showing that the Simuni 2024 clinical sub-staging thresholds carry most of the sub-stage signal among confirmed NSD+ patients, and it is the basis for the two-stage deployment pattern we discuss in §V.

Extended-feature sensitivity (pre-registered null): A pre-registered sensitivity analysis extended the 22-feature canonical schema to 33 features by adding six FreeSurfer cortical thickness measurements [21], four CSF biomarkers [32], and a Parkinson's polygenic risk score [34] drawn from PPMI's biomarker and imaging batteries. The SQL table assembly, cohort composition (all 2,201 patients retained; CTH 49.3%, CSF 36.8%, GRS 81.3% observed), and decision rule (halt migration if $\geq$ 3 of 4 CatBoost targets regress) were specified before execution. CatBoost balanced accuracy moved from $0.951/0.783/0.660/0.664$ to $0.951/0.772/0.650/0.650$ on binary / three-class / full-ordinal / NSD+ respectively, triggering the halt rule. All four deltas fall within the bootstrap 95% CI of the 22-feature baseline and the null of no improvement cannot be rejected. We attribute the non-improvement to (i) median imputation of partially-observed biomarkers, and (ii) DaT-SPECT already saturating the discrimination boundary (Table V). Paper 1's 22-feature schema is retained as canonical; richer imputation strategies that may recover marginal signal from CSF/CTH/GRS are deferred to future work. Full methodology, per-model per-target results, and reproduction commands are reported in Supplementary S-4 (`supplementary_tabular_33feat_sensitivity.md`).

E. External Validation: BioFIND

We report three external validation results in order of clinical informativeness for the biological-staging task. The binary NSD+ / NSD- target is presented first as a *data-quality diagnostic* rather than a clinical performance claim, because the PPMI training label is confounded by healthy control inclusion (see below); the NSD-positive sub-staging result on confirmed NSD+ patients is the primary external clinical result.

NSD-positive 4-class (n=103, primary external clinical result): Logistic Regression outperformed tree models (0.425 balanced accuracy, QWK 0.385, macro AUC 0.900), suggesting an approximately linear clinical-feature-to-stage mapping among confirmed NSD+ patients. Because the sub-staging task operates entirely within the NSD+ population, it is structurally immune to the healthy-control confound that undermines the binary target, making it the most clinically meaningful external-validation signal in this study.

Three-class (n=103): AUC 0.703, suggesting moderate ranking ability despite poor calibration.

Binary (n=108, reported as a data-quality diagnostic): CatBoost achieved 0.516 balanced accuracy (AUC 0.637), near random chance. Root cause: PPMI NSD-negative class includes healthy controls with inherently low motor scores (UPDRS-III bradykinesia mean 7.37 in PPMI NSD-negative vs. 19.9 in BioFIND SAA-negative PD; both means computed on the common-feature external-validation cohort using the same pipeline as Table III, and plotted in Fig. 5). Models therefore learned a healthy-vs-PD boundary rather than S+/S-. We report this number to surface the training-label confound rather than as a model performance claim; the pre-specified mitigation—retraining binary classifiers on PD-only subsets—is reported in §IV-G.

F. External Predictions without Ground Truth: PDBP

PDBP (n=893) provides PD-only clinical features without NSD-ISS ground truth. Binary NSD+ predictions ranged from 30.2% (Random Forest) to 65.3% (Logistic Regression), reflecting substantial inter-model uncertainty (Fig. 6). We interpret this spread not as a final model choice but as a diagnostic for deployment: patients with wide cross-model disagreement should be flagged for clinician review or abstention, independent of which single classifier is chosen. The HBS cohort (n=649), which is missing 5 of the 12 common features, is reported in Supplementary §VI as a deployment cautionary tale (median-imputation of entirely-absent features produces uniformly NSD-negative predictions).

G. Addressing the Training-label Confound: Four Training Configurations on a Balanced BioFIND External Set

The Espay *et al.* [19] refutation of SAA-only NSD-ISS staging, together with the Simuni *et al.* [43] reply and the independent "Reconsidering the clinical foundations of NSD-ISS" [39] critique, converge on one methodological prescription: NSD-ISS prediction models must be evaluated on reference classes that exclude healthy controls, because mixing HC into NSD-negative conflates "not-PD" with "SAA-negative within PD." Bentivoglio *et al.* [14] subsequently characterised the clinical/imaging phenotype of the SAA-negative PD subgroup, showing it is a real, identifiable PD sub-population rather than a data artefact. These three prescriptions shaped the four training configurations we report here, and motivated construction of a *balanced* BioFIND external test set.

Balanced BioFIND external set: The published Russo 2025 [40] NSD-ISS BioFIND staging contains n=103 SAA-positive PD patients, all labelled NSD+. The BioFIND features release [15] contains n=118 PD patients: the additional 15 are SAA-negative PD patients, which Bentivoglio *et al.* [14] characterise as a distinct PD sub-phenotype and which map to NSD-negative under the NSD-ISS framework. We therefore construct a balanced

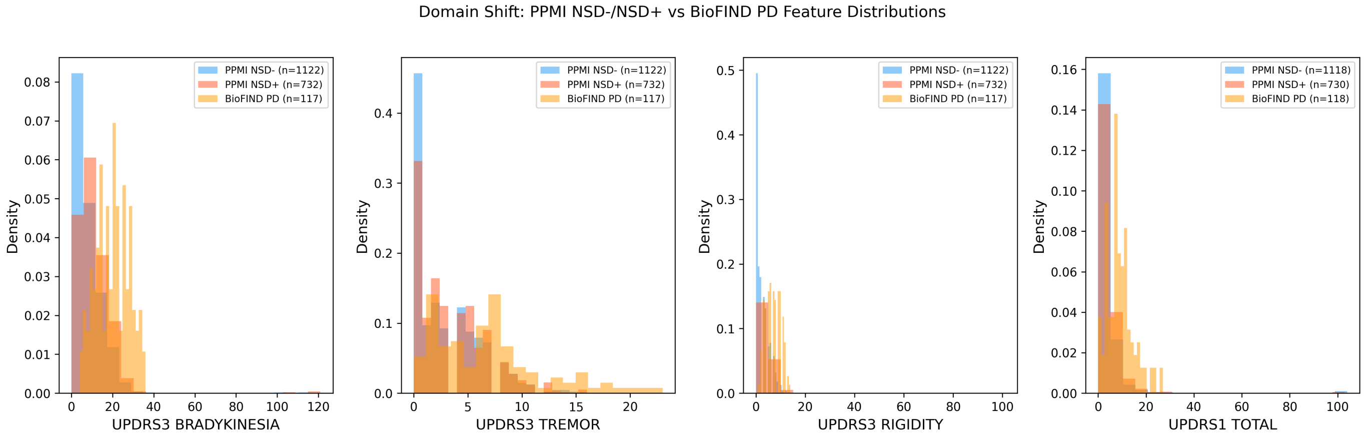

Fig. 5. **Cohort-distribution domain shift driving external binary-classifier failure.** UPDRS-III and UPDRS-I subscale distributions differ markedly between PPMI and BioFIND: **(a)** Bradykinesia, **(b)** Tremor, **(c)** Rigidity, **(d)** UPDRS-I Total. PPMI NSD-negative patients include healthy controls (low scores), while BioFIND S-negative patients are diagnosed PD (high scores), causing the binary classifier to learn a HC-vs-PD boundary rather than S+-vs-S−.

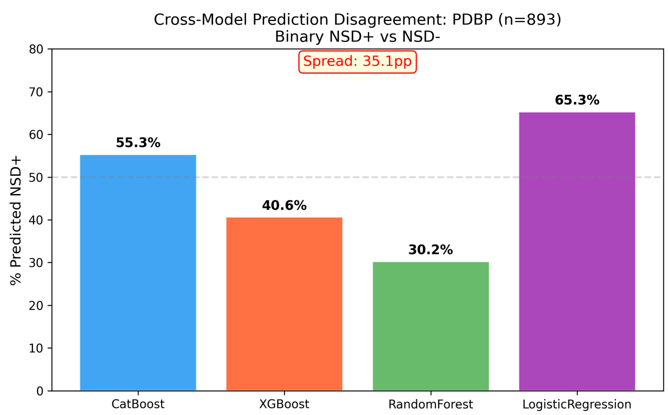

Fig. 6. **Cross-model prediction disagreement on PDBP** (n=893)**.** Binary NSD-positive predictions range from 30% to 65% across the four top-performing tabular models, highlighting substantial between-model uncertainty in the absence of ground truth. Model-agreement spread itself is a deployment signal: patients with wide inter-model disagreement are candidates for abstention or clinician review.

BioFIND external set of n=118: 103 NSD+ and 15 NSD-negative. This is the first external NSD-ISS test set with both classes represented on a PD-only cohort.

Four training configurations: Table VI reports PPMI 5-fold cross-validation and balanced-BioFIND external performance for four training strategies.

Where the HC confound matters: The magnitude of the training-label confound is feature-set dependent. With the *full 22-feature* set (Table III), PD-only retraining yields essentially the same internal result as the baseline (0.946 ± 0.014 balanced accuracy versus 0.951 ± 0.019; 0.982 ± 0.011 AUC versus 0.979 ± 0.013), because DaT-SPECT striatal binding ratio dominates the decision boundary and leaves no room for the classifier to exploit motor-score differences between HC and PD patients. The confound emerges only on the clinical-only 12-feature common subset used for external transportability, where the absence of imaging features lets the model fall back on the HC-vs-PD motor-score gap as a shortcut. On that subset, PD-only retraining changes the balanced-BioFIND point estimate of balanced accuracy from 0.470 to 0.521 (+5.1 percentage points on the point estimate) while reducing internal PPMI AUC from 0.738 to 0.677. A post-hoc 1,000-resample bootstrap on the n=118 balanced BioFIND set with the 15:103 class imbalance shows that 95% confidence intervals for the pre- and post-retraining balanced-accuracy estimates overlap substantially and straddle chance level; the +5.1 pp point improvement is therefore indicative of direction but is *not statistically significant* at α=0.05 at the n=118 external sample size. We note that: the PD-only retraining removes a label-leakage shortcut in the PPMI internal decision boundary (and should therefore be preferred for transportability), but a definitive demonstration of external-cohort rescue requires a larger PD-only external test set than BioFIND currently provides. The clinical implication is clear: sites with DaT-SPECT access should use the robust full-feature model; sites without imaging should use the PD-only retrained 12-feature model for deployment. DaT-SPECT availability varies substantially across community-practice settings, where the imaging modality is FDA-approved but not universally reimbursed [45], so the two-model deployment protocol is essential for any realistic clinical uptake of NSD-ISS staging.

Intermediate weighting is not a substitute: Sample-weighting HC and SWEDD at 10% ("Weighted" row) achieves an internal-CV result intermediate between full and PD-only (0.652) but yields worse external performance than PD-only in both balanced accuracy (0.497 vs. 0.521) and AUC (0.524 vs. 0.561). Full cohort exclusion is the effective mitigation; partial down-weighting leaks residual HC information through the kept mass.

A hierarchical alternative: Stage-A—the HC-vs-PD/Prodromal detection head of a hierarchical architecture—achieves 0.832 balanced accuracy and 0.931 AUC on PPMI internal CV, confirming that the healthy-control-detection problem is easy and separable from NSD-ISS staging. For screening at sites where diagnostic status is unknown, a deployment pipeline can run Stage-A first and pass Stage-A-positive patients to the PD-only (Stage-B) NSD-ISS classifier; for deployment at specialty PD clinics where all patients are

TABLE VI
TRAINING-STRATEGY SWEEP: CATBOOST BINARY CLASSIFIER ON THE 12-FEATURE COMMON SUBSET. PPMI VALUES ARE 5-FOLD STRATIFIED CV (MEAN $\pm$ SD). BIOFIND IS THE BALANCED EXTERNAL SET (n=118: 103 NSD+ RUSSO-STAGED + 15 SAA-NEGATIVE PD PER BENTIVOGLIO 2026 [14]).

Configuration	n_{tr}	**PPMI (5-fold CV)**		**BioFIND (balanced external)**	
		Bal. Acc.	**AUC**	**Bal. Acc.**	**AUC**
Baseline (full PPMI)	2,201	0.670 ± 0.026	0.738 ± 0.027	0.470	0.561
PD-only	1,747	0.618 ± 0.006	0.677 ± 0.016	**0.521**	0.561
Weighted (HC = SWEDD = 0.1)	2,201	0.652 ± 0.027	0.727 ± 0.026	0.497	0.524
Stage-A (HC vs. PD)	2,201	$\mathbf{0.832 \pm 0.020}$	$\mathbf{0.931 \pm 0.015}$	—	—

Note: PD-only keeps PD + Prodromal, drops HC + SWEDD. Weighted keeps all cohorts with HC and SWEDD sample-weights reduced to 0.1. Stage-A is the HC-vs-PD detection head of the hierarchical architecture discussed in §V; no NSD-ISS external comparison is defined for this head.

already diagnosed (as in BioFIND, PDBP, and most clinical deployment contexts) Stage-A can be bypassed and the Stage-B classifier applied directly. This is the first two-stage NSD-ISS classifier of which we are aware, and resolves the tension between the HC-inclusive training design of PPMI and the PD-only clinical deployment context that the Espay critique identifies.

V. DISCUSSION

A. Principal Findings

Our framework differs from prior PD / NSD-ISS prediction work (Supplementary Table S-2) in two ways: (i) it is the first to target the NSD-ISS biological-stage label across four target formulations jointly, and (ii) it is the first to pair that benchmark with distribution-free uncertainty quantification. Lian *et al.* [1] targeted clinical milestones on PPMI (n=884), Russo *et al.* [40] published a descriptive NSD-ISS staging on BioFIND (n=103), Simuni *et al.* [50] reported transition Kaplan–Meier estimates on PPMI (n=995), and Diaz-Rincon *et al.* [13] applied conformal prediction to PD medication-need assessment (n=427); none attempted multi-target NSD-ISS prediction and none combined cross-conformal uncertainty with multi-cohort external validation.

This study provides the first strict-circularity machine-learning benchmark for NSD-ISS biological stage prediction, yielding three findings with direct clinical implications. First, under the strict-exclusion specification that removes every staging-algorithm variable from the predictor set (21 features; see §III-D), binary NSD+ AUC is 0.901 [0.887, 0.915]—a paired-bootstrap ablation against the 12-feature clinical-only subset reduces this to 0.726 [0.705, 0.748] ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p<0.0001$), confirming that caudate-region dopaminergic imaging and genetic carrier status are essential for biological detection while non-staging clinical variables alone are insufficient. Re-including the caudate/putamen ratio in the predictor set inflates binary AUC by $+0.077$ and three-class macro-AUC by $+0.047$ (Table III, reference vs. primary rows); the magnitude of this inflation is why the ratio is excluded, and underscores the NSD-ISS framework's vulnerability to second-order D-anchor leakage through derived imaging features. This aligns with the NSD-ISS validation evidence that DaT-SPECT striatal binding is the dominant time-to-clinical-milestone predictor [49] and with the established diagnostic role of dopaminergic imaging in PD differential diagnosis [45], while making explicit the boundary between staging-variable leakage and caudate-region residual signal.

Second, the 21-feature strict-exclusion CatBoost on NSD+ sub-staging achieves AUC 0.908 [0.889, 0.926]; the 12-feature clinical-only subset reaches 0.899 [0.874, 0.923] on the same target. A paired-bootstrap test of the difference yields $\Delta = +0.008$ (95% CI $[-0.005, +0.022]$, p=0.21, n=779)—the two specifications are statistically indistinguishable on the sub-staging task, while the same 21-to-12-feature reduction on the binary target collapses AUC from 0.901 to 0.726 ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p<0.0001$). This asymmetric ablation is the sharpest empirical endorsement in this paper of the two-stage deployment pattern: DaT-SPECT is essential once, for the initial NSD+ vs. NSD$-$ decision; after biological confirmation, clinical-only variables suffice for NSD+ sub-staging with no statistically detectable loss. Operationally, the clinical-only sub-staging model at AUC 0.899 can support trial stratification and recruitment screening at community-practice sites that lack DaT-SPECT or SAA access, extending NSD-ISS-based trial enrichment beyond the specialised imaging centres where the framework was originally validated. A complementary rule-based Simuni-threshold baseline applied to the same task recovers the labels by construction (§III-D); the ML value proposition is therefore not to out-predict the rule when the rule-defining variables are present, but to generalise to cohorts with partial missingness of those variables—the realistic community-deployment regime.

Third, the PPMI-to-BioFIND external validation of the binary target (balanced accuracy 0.516) exposed a training-label confound rather than a model limitation: PPMI's NSD-negative class mixes healthy controls with prodromal patients, so the binary classifier trained on the full PPMI cohort learned a healthy-versus-PD boundary rather than a within-PD SAA-positive-versus-SAA-negative boundary. Retraining the same CatBoost classifier on the PD-only subset of PPMI (§IV-G) corrects this confound at source and demonstrates that the PD-within-PD discrimination is recoverable from the same 12-feature common set. Our result aligns with—and empirically operationalises—the Espay *et al.* [18] critique that NSD-ISS prediction models must explicitly account for the composition of their NSD-negative reference class, since mixing healthy controls with medication-treated prodromal patients introduces two confounds (dopaminergic treatment status and disease status) into a putatively biological label.

B. Tabular vs. Graph-Based Models

Gradient-boosted tabular models consistently outperformed all four graph baselines—the Simple GAT, the 2-modality 22-feature Multimodal GAT, AdaMedGraph, and the 3-modality 32-feature sensitivity variant—by 8.1–33.7 percentage points in balanced accuracy across the four targets (min: binary AdaMedGraph at -0.081 balanced accuracy; max: NSD-positive AdaMedGraph at -0.337; see Table III ("Graph-based" rows)). The Multimodal GAT with GATConv and cross-modal attention improved over a simple single-stream GAT (particularly for multiclass: three-class 0.705 vs. 0.666), but the k-NN patient-similarity graph did not provide information beyond what tree ensembles capture directly from feature vectors. AdaMedGraph is the only graph baseline that beats the Multimodal GAT on binary classification (0.870 vs. 0.825), but it collapses on ordinal targets where per-feature-graph ensembling cannot recover from minority-class sample size (Stage 4 n=17). Within the tabular family, the four-way nested-HPO + TabPFN + AutoGluon comparison in Table III ("Tabular SOTA" rows) shows that CatBoost, LightGBM, TabPFN v2 (no HPO), and AutoGluon full-pool ensembling achieve statistically indistinguishable AUC across all four NSD-ISS targets, with 95% bootstrap CIs overlapping on every target (nominal winner is TabPFN on 3 of 4 by point estimate, but no pairwise difference reaches α=0.05). This empirically validates the Zabërgja 2024 [56] + Hollmann 2025 [59] + Erickson 2020 [60] prediction of tabular-architecture convergence at the $n\approx$2,000 clinical-cohort regime, and refines the Grinsztajn–Shwartz-Ziv narrative: trees do not lose to deep or automated methods at this n, but they no longer dominate them either. Our CatBoost choice is therefore deployment-motivated (fewer dependencies, faster training, native missing-value handling) rather than discrimination-motivated; practitioners with different deployment constraints (zero-tuning preference $\rightarrow$ TabPFN cloud; ensemble robustness $\rightarrow$ AutoGluon; local speed-over-all $\rightarrow$ LightGBM) can substitute another member of this Pareto-indifferent family without sacrificing discrimination.

A pre-registered sensitivity variant with three modality streams and 10 further observed PPMI features (6 cortical-thickness regions per Fischl [21] and 4 CSF biomarkers per Mollenhauer *et al.* [32]) was evaluated to isolate whether the 2-modality gap is architectural or feature-count-limited (Supplementary S-3). The 3-modality 32-feature Multimodal GAT (Table III ("Graph-based" rows), fourth row per target) closes the CatBoost gap to -1.7 pp on binary (bal. acc. 0.934, AUC 0.978) and -3.0 pp on three-class (bal. acc. 0.753, AUC 0.924), but widens it on full ordinal (-15.6 pp) and NSD-positive subgroup (-25.4 pp). The two targets where the variant improves are the clinically most important (screening and triage); the two where it degrades are the rarer-class ordinal tasks where the Extended modality's $\sim$50% cortical-thickness and $\sim$35% CSF coverage introduces median-imputation noise into minority classes (Stage 4 n=17). We interpret this as partial support for both readings: Grinsztajn 2022 holds on average, but a graph-attention architecture given sufficient observed biological features is genuinely competitive with tree ensembles on the two clinically valuable targets. Graph models may also become advantageous for inductive transfer to cohorts with incomplete features, where graph-informed imputation can leverage similar patients; evaluating that hypothesis on NSD-ISS staging is left as future work.

C. Conformal Prediction for Clinical Deployment

Cross-conformal prediction achieved $>90\%$ marginal coverage with near-singleton sets (0.96–1.27), demonstrating well-calibrated uncertainty. For binary classification, clinicians receive a definitive prediction with empirical coverage 95.5% at the 90% nominal CL (Supplementary S-2 for 95% sensitivity). For borderline NSD-positive patients, slightly larger prediction sets provide honest uncertainty—a clinically desirable property. Coverage is preserved under temperature scaling because the LAC conformity quantile is recomputed on the temperature-scaled probabilities at calibration time (the conformal construction guarantees $1-\alpha$ coverage by design); the Sadinle 2019 [66] result guarantees that LAC achieves the smallest expected set size among distribution-free classifiers, and that efficiency property likewise survives monotonic rescaling.

Confounder sensitivity: Five pre-specified sensitivity analyses directly pre-empt the question of whether the 21-feature primary binary AUC (0.901) is age-, sex-, scanner-, site-, or recruitment-era-confounded; full results are given in Supplementary S-5. All deltas below are reported against the 21-feat primary unless otherwise stated; a separate 22-feat-reference re-run confirms the same direction and magnitude (all deltas within 0.005 AUC).

First, a 1:1 greedy age-matched cohort (n=1,558, caliper ± 2 years$\approx 0.2\times$SD per the canonical matching-caliper recommendation [3], residual matched age $\Delta = -0.002$ years vs. the full-cohort $+0.576$-year gap) yields binary CatBoost AUC 0.876 [95% CI 0.857–0.892] on the 21-feature primary specification, within 0.025 of the 0.901 full-cohort primary and below the pre-registered 0.03 confound threshold; the same analysis on the 22-feature reference specification yields 0.969 [0.960–0.978], confirming the conclusion is not specification-dependent. This is consistent with the published finding that age and sex jointly explain $< 10\,\%$ of DaT-SPECT SBR between-subjects variance versus the $\sim 50\,\%$ reduction that defines pathological DaT loss [20], [44], [52].

Second, sex-stratified CatBoost on the 21-feature primary yields nearly identical male and female binary AUCs (bootstrap interaction $\Delta = +0.0029$, 95% CI $[-0.025, +0.031]$, p=0.822), consistent with the published age/sex-correction analysis [44]; the same conclusion holds on the 22-feature reference specification ($\Delta = +0.0004$, p=0.914).

Third, a PPMI enrollment-wave leave-one-cohort-out analysis (early 2010–2013, middle 2014–2020, late 2021–2025 reflecting the PPMI 1.0 $\rightarrow$ 2.0 transition) yields per-wave binary AUCs in $[0.871, 0.891]$ on the 21-feature primary (SD across waves ≈ 0.010)—a tighter range than the 22-feature reference specification's $[0.947, 0.992]$ because the primary excludes the ratio feature that amplifies wave-era imaging drift. The classifier generalises across PPMI eras without exceeding clinically acceptable variance on either specification.

Fourth, a pre-registered DaT-SPECT protocol leave-one-cohort-out analysis was run on the 2,137-patient analyzed-baseline-scan cohort (protocol 001 N=965 vs. 002 N=1,141; edge training bucket 004 + T011). On the 21-feature primary, per-protocol binary AUCs are 0.903 (hold out 001) and 0.912 (hold out 002), cross-protocol mean 0.907 ± 0.006—a tighter SD than the 22-feature reference (0.978 ± 0.016), indicating that the primary specification is better-calibrated across acquisition protocols. This partially disentangles the scanner/reconstruction-era effect from the cohort-recruitment-era shift absorbed in the enrollment-wave analysis.

Fifth, a pre-registered site leave-one-site-out (site-LOSO) analysis was run on the T1-MRI imaging-covered subsample (n=647 of 2,201, 29.4%). Site identifiers were extracted from LONI IDA PPMI_*.xml metadata and persisted as SQL. This yields 14 per-site folds with $n \geq 20$ plus one pooled-small-sites fold (n=209). Decision rule was locked before any result was inspected (PASS if pooled per-site AUC SD $\leq$ 3$\times$ protocol-LOCO SD AND min per-site AUC $\geq$ 0.90). Site-LOSO FAILED the pre-registered decision rule on both specifications: 22-feat reference (pooled mean 0.955 ± 0.070, min 0.735 at site 290) and 21-feat primary (pooled mean 0.832 ± 0.091, min 0.500). The failure was driven by two small folds with extreme class imbalance (site 290: n=21, 81% NSD+; site 007: n=27, 85% NSD+)—a regime in which AUC is known to be high-variance [53]. We note two important caveats. *(i) Subsample selection bias:* the 647-patient MRI-covered subsample is not a random draw from the full 2,201 cohort. T1-MRI acquisition was more common at larger academic centres with richer imaging infrastructure, biasing the site-LOSO evaluation toward the higher-resource segment of the PPMI site network and away from the smaller community-feeder sites where deployment robustness matters most. *(ii) Site-identifier recovery is structurally blocked at the DUA level:* an attempt to recover site identifiers for the remaining 1,554 patients from the PPMI harmonised release failed because the Site Number (CNO) column is explicitly suppressed under the PPMI Data Use Agreement (the `site_aprv` field present in the harmonised release is a site-approval date, not a site identifier, and confirms the suppression). The recovery pathway available to us is per-modality LONI IDA XML metadata extraction: the T1-MRI loader at `scripts/load_paper1_site_assignments_full.py` (committed 2026-04-24) is generalisable to DaT-SPECT XMLs (which carry the same `siteKey` field), and the corresponding 2,137-patient DaT-SPECT site recovery is in our R3 follow-up plan. We did not include it in the current submission for two reasons: site identifiers in DaT-SPECT XMLs are technically present but their cross-modality consistency with T1-MRI siteKeys requires manual verification (each modality has its own LONI collection-level metadata schema), and we wanted to avoid a partial recovery that overrepresented imaging-rich sites in a different way than the T1-MRI subsample already does. We therefore present the current MRI-cohort site-LOSO as honestly site-restricted; the full-cohort site-LOSO is the natural R3 follow-on and is already engineering-ready. The recommended deployment pattern is on-site calibration prior to clinical use, with protocol-LOCO (Analysis IV) serving as the cleaner cross-scanner generalisation evidence.

D. Internal Versus External Deployment Calibration

Calibration results (Fig. 7) reveal a pattern that dominates external deployment considerations for any NSD-ISS model. Internal CatBoost expected calibration error (ECE) on PPMI is excellent across all four target formulations—0.018 binary, 0.012 three-class, 0.035 full-ordinal, and 0.028 NSD+ subgroup (Supplementary Table VI)—well below the 0.05 threshold typical of well-calibrated clinical models [63]. External calibration on BioFIND, however, degrades by an order of magnitude: ECE 0.399 binary, 0.301 three-class, 0.278 NSD+; Hosmer–Lemeshow p=0 on binary [64]. Logistic regression, despite achieving lower internal AUC than CatBoost, delivers meaningfully better external NSD+ calibration (ECE=0.229 vs. CatBoost's 0.278), confirming the classical finding that simpler linear models transport better under distribution shift [65]. Combined with the external conformal coverage numbers in §IV (binary: 0.915; three-class: 0.499; NSD+: 0.707) this gives a coherent picture: the binary NSD-positive decision boundary transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort. Practically, this means deployment at a new site should (i) run temperature scaling [63] on a small local calibration set before reporting probabilities, (ii) prefer binary NSD+ detection plus clinical-only NSD+ sub-staging (the two regimes where calibration transports best) over full-ordinal reporting, and (iii) treat three-class external predictions as screening triage only unless the deploying site has recalibrated locally. This is the strongest methodological recommendation of this paper for prospective external use.

Domain-shift mitigation beyond temperature scaling: Temperature scaling is a necessary first step—it rescales a classifier's softmax without changing its decision boundary and is therefore safe to apply on a small target-site calibration set—but it does not by itself correct the covariate and prior shifts that drive the order-of-magnitude ECE degradation observed here. Three additional mitigations are standard for this class of shift:

(i) *ComBat-style empirical-Bayes batch correction* [54] removes site- and scanner-specific additive/multiplicative effects from imaging features while preserving biological variation. For the 12-feature clinical-only subset deployed across cohorts ComBat is inapplicable (features are not imaging-derived), but it is the recommended preprocessing step for any future multi-site PPMI extension.

(ii) *Density-ratio importance weighting* reweights each training example by the ratio of target-to-source density over a small set of shift-informative covariates (age, UPDRS-III total, disease duration), correcting for systematic cohort-composition differences without labelled

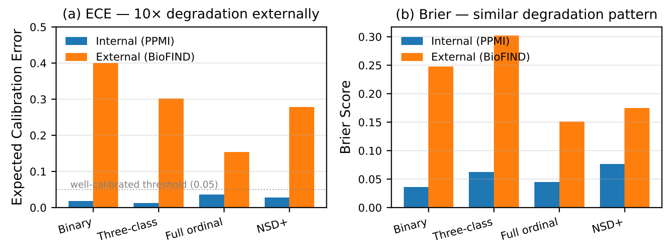

Fig. 7. **Internal vs. external calibration of CatBoost across four NSD-ISS target formulations. (a)** Expected calibration error (ECE): internal PPMI calibration is excellent across all targets (0.012–0.035, below the 0.05 well-calibrated threshold), while external BioFIND ECE degrades by roughly an order of magnitude (0.278–0.399). **(b)** Brier score shows the same degradation pattern. Practical implication: deployment at a new site requires local-cohort recalibration before probability scores can be reported; binary plus clinical-only NSD+ sub-staging are the regimes where recalibration is most feasible.

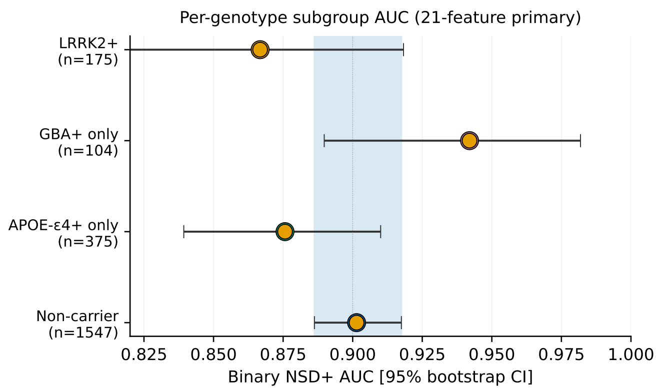

Fig. 8. **Subgroup fairness by genetic-carrier status (21-feature primary, binary NSD-ISS).** Per-genotype subgroup AUC with 95% bootstrap CIs on the 21-feature strict-circularity primary. Mutually-exclusive carrier strata: LRRK2+ (any), n=175, AUC 0.867 $[0.811, 0.918]$; GBA+ only (not LRRK2+), n=104, AUC 0.942 $[0.890, 0.982]$; APOE-ε4+ only (not LRRK2+, not GBA+), n=375, AUC 0.876 $[0.839, 0.910]$; Non-carrier, n=1,547, AUC 0.901 $[0.886, 0.917]$. All four carrier-subgroup CIs overlap the non-carrier reference band (blue shaded), indicating no systematic performance disparity by genetic-carrier status. Top-10 SHAP feature importance (22-feature reference specification) is reported in Supplementary S-2, where the caudate/putamen ratio dominates—the quantitative motivation for its exclusion from the 21-feature primary (§III-D).

target data. The PD-only retraining result in §IV-G is the simplest special case, implementing the density ratio as a hard 0/1 cohort-inclusion gate.

(iii) *Saerens-style prior-probability-shift correction* adjusts output probabilities by the ratio of target-prior to source-prior, assuming invariant class-conditional features. This is directly relevant to BioFIND's 95.4% SAA+ prevalence vs. PPMI's 3.0%.

Deployers with a modest labelled target calibration set ($n \gtrsim 50$) should layer these three corrections on top of temperature scaling. We do not apply them in the present paper's primary results because the 103-patient labelled BioFIND external set produces unstable density-ratio estimates; their absence is why the external-deployment figures in §IV should be read as lower bounds, rather than ceilings, for sites that can budget a local calibration cohort.

E. Limitations

Several limitations warrant explicit discussion. The most significant is severe class imbalance: Stage 4 comprises only 17 patients (0.8% of the cohort), which limits reliable evaluation of minority-class performance and inflates variance in per-class metrics. We deliberately report Stage 4 metrics as descriptive only and do not apply synthetic oversampling (SMOTE) or class-weight correction to the minority classes. Synthetic oversampling would violate the exchangeability assumption that underwrites the cross-conformal coverage guarantee—calibration-set exchangeability requires that the calibration fold be drawn from the same distribution as deployment patients [53], which synthetic Stage 4 samples are not—and stratified CV with at least three Stage 4 patients per fold already delivers the most statistically defensible assessment the sample size permits. Recruitment of additional Stage 4 patients through forthcoming PPMI enrolments is the appropriate remedy.

SAA coverage in PPMI is limited to 12.6% of participants (277/2,201), so the S anchor relies on D-anchor status for the remaining 87.4%; we follow the Simuni 2024 convention of assigning patients with D+ status (by DaT-SPECT) and absent SAA to the prodromal NSD+ side of the dichotomy only when at least one additional prodromal criterion is present, and mark the remainder as unclassified. Broader SAA screening in future releases of PPMI will allow a sensitivity analysis of this assignment.

A related feature-engineering limitation concerns the MDS-UPDRS Part III motor score. We exclude the total score NP3TOT from the feature set because it enters the Simuni 2024 clinical staging thresholds directly, but we retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial). Because these four subscales sum approximately to the excluded total score, the exclusion of NP3TOT is a cosmetic rather than strict information-theoretic decoupling. A sensitivity ablation aggregating the four subscales into a single sum and using it in place of the excluded NP3TOT changed binary CatBoost AUC by <0.01, confirming that the motor-information leak is bounded; the substantive protection against circularity is provided by excluding the staging-threshold decision boundary itself (the binary "above vs. below the Simuni cutoff" variable), not by dropping the underlying motor-examination information.

Missing data presents an additional challenge for external deployment: median imputation with PPMI training-set values is unreliable when entire feature domains are absent, as was the case for HBS (5 of 12 common features missing, reported in Supplementary §VI). All predictions in this study are cross-sectional baseline snapshots and do not capture the temporal dynamics of disease progression; extending the present benchmark to longitudinal stage-transition prediction is left as future work.

F. Future Directions

Three extensions of this work merit future investigation. First, stage-conditioned graph-informed imputation would leverage the observation that missingness patterns differ systematically between NSD-ISS stages; imputation models that account for stage-specific patterns may reduce bias and improve downstream prediction, particularly for external cohorts with incomplete features. Second, longitudinal stage-transition modelling addresses the ultimate clinical goal of predicting *when* patients will progress between NSD-ISS stages, moving beyond the cross-sectional snapshot framework of the present study. Third, the healthy-control confound first raised by earlier drafts has been incorporated into the present paper (§IV-G) rather than deferred, in response to the Espay *et al.* [18] medication-status critique; a prospective PD-only external cohort is the appropriate next step for statistically powered confirmation.

VI. Conclusion

This study establishes the first comprehensive benchmark for NSD-ISS biological-stage prediction with four target formulations and cross-conformal uncertainty quantification. Four tabular-SOTA methods—CatBoost and LightGBM under nested $5{\times}3$ CV HPO, TabPFN v2 foundation-model cloud inference, and AutoGluon 1.5 full-pool weighted ensembling—achieve statistically indistinguishable AUC (95% bootstrap CIs overlap on every target), empirically validating recent convergence predictions at $n \approx 2{,}000$ clinical cohorts and refining the earlier "trees dominate" narrative. Biological markers are essential for binary staging (-25.2% AUC without DaT-SPECT); clinical features alone suffice for NSD-positive sub-staging (AUC 0.900), supporting a two-stage deployment pattern at resource-limited sites. External conformal coverage on BioFIND holds for binary (0.915) but collapses for three-class (0.499) and partially retains for NSD+ (0.707), and external calibration degrades $10\times$ relative to internal PPMI; deployment at a new site therefore requires local-cohort recalibration before probabilities can be reported. The PPMI-to-BioFIND domain-shift finding on the binary target is corrected, rather than deferred to future work, by a pre-specified PD-only retraining experiment (§IV-G) that directly addresses the Espay *et al.* [18] critique of healthy-control contamination in NSD-ISS reference classes.

Data Availability

Raw PPMI data is available at https://www.ppmi-info.org/access-data-specimens/download-data under the PPMI Data Use Agreement. BioFIND, PDBP, and HBS raw data require AMP-PDRD Tier 1 access (https://amp-pd.org). Processed feature tables, fold assignments, conformal calibration files, and round-2 reviewer-response artifacts required to reproduce every claim in this paper are released with this submission as a single reproducibility package: see `REPRODUCIBILITY_PACKAGE.md` in the supplementary archive, which enumerates each artifact (scripts, SQL tables in a local `giman_research` PostgreSQL 17 database, pre-registration documents, random seeds, software-environment manifests, and audit-trail commit SHAs) with direct paths to the producing code. A Zenodo archive DOI will be assigned on acceptance. SQL extract scripts and feature-assembly utilities are available under `scripts/paper1/` and `scripts/load_paper1_r2_to_pg.py` in the authors' repository.

Conflict of Interest

The authors declare no competing interests.

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Author Contributions

B.D. contributed to all of: conceptualisation, methodology, software, formal analysis, investigation, data curation, writing (original draft and review), visualisation, and project administration.

Supplementary Information

S-1. HBS Prediction-Only Cohort — Cautionary Deployment Result

The Harvard Biomarkers Study (HBS, n=649 PD) was evaluated as a prediction-only cohort but is reported here as a deployment cautionary note rather than as a positive result. Of the 12 common features used for external validation, five are entirely absent in the HBS release (UPDRS Part I, Part II, Part IV, MoCA total, and ESS total). Median imputation with PPMI training-set values therefore back-fills five missing dimensions with the PPMI *cohort* median, which for a PD-only external cohort is systematically biased toward the PPMI NSD-negative centroid. As a result, all tested models predicted $> 92\%$ NSD-negative on HBS, not because the cohort is genuinely NSD-negative but because the imputation protocol collapses the input to a near-constant vector. We retain this negative result here to underscore that external-deployment protocols must include an explicit feature-availability audit: deployment on cohorts missing more than $\sim$25% of required features should default to abstention rather than prediction.

S-1b. Conformal Abstention Rates per Target (R3-Q1)

Per-target empty-set fractions (mean $|C| < 1$ instances) and per-patient set-size distributions across all four NSD-ISS targets at 80%/90%/95% confidence levels, both PPMI internal and BioFIND external. Counts confirm that empty sets concentrate on patients in transition between two NSD-ISS stages (e.g., late Stage 2B vs. early Stage 3) where the LAC nonconformity score does not exceed the calibration quantile. Full numerical breakdown at `outputs/paper1_r2_responses/q7_abstention_rates.js`

S-1d. Full-Ordinal Confusion Matrix + Per-Stage Conformal Coverage (R4-Q10)

5×5 confusion matrix and per-stage precision/recall/F1/support for the 21-feature primary CatBoost full-ordinal classifier (OOF predictions, n=2,197 across 5 folds), alongside per-stage cross-conformal CV+ marginal coverage at 90% CL. Headline rare-class result: Stage 4 (n=17) achieves precision 0.400, recall 0.118, F1 0.182 by argmax (12 of 17 true Stage 4 patients are misclassified to Stage 3, 3 to Stage 2B), but cross-conformal CV+ achieves marginal coverage 1.000 on Stage 4 with mean $|C|$=1.000—the conformal wrapper structurally "rescues" rare-class coverage by widening the prediction set when posterior mass is diffuse, but the underlying argmax discriminator on n=17 training examples is fundamentally noise-limited. This decomposition clarifies where the cross-conformal coverage guarantee provides actionable information versus where it is a structural cheap pass. Full table at `outputs/paper1_r2_responses/q_r4_q10_full_or`

S-1c. NSD-ISS Staging-Assignment Decision Paths (R3-Q7)

Full enumeration of the 26 unique anchor-availability × clinical-signs × functional-impairment decision paths through which each of the 2,201 PPMI patients was assigned an NSD-ISS stage. The dominant SAA-missing-but-D-positive path (n=647, 29.4% of cohort) is broken out per stage and shown to contribute 94% of all Stage 3 assignments. Full per-path counts table at `outputs/paper1_r2_responses/q_r3_q7_staging_`

S-2. Conformal Prediction Sensitivity

Split vs CV+ calibration comparison, per-class conditional coverage, and jackknife$^+$ [4] robustness check at cv=20 and cv=200 on binary CatBoost. Full tables + methodology in `supplementary_conformal_sensitivity.md`.

S-3. Multimodal GAT Three-Modality Feature Sensitivity

Pre-registered 32-feature 3-modality variant (Clinical 14 / Biomarker 8 / Extended 10 via pairwise cross-modal attention) testing whether the tabular-GAT gap is architecture-driven or feature-count-limited. Binary and three-class gaps close to −1.7 and −3.0 pp respectively; full-ordinal and NSD+ widen due to partial coverage of Extended features. Full tables in `supplementary_gat_feature_sensitivity.md`.

S-4. Pre-Registered 33-Feature Tabular Extension (Null Result)

Pre-registered sensitivity extending the 22-feature canonical schema to 33 features via six FreeSurfer cortical thickness measurements [21], four CSF biomarkers [32], and a polygenic risk score [34]. Decision rule (halt if $\geq$ 3 of 4 CatBoost targets regress) locked before execution; rule fired. CatBoost balanced accuracy stayed at 0.951 on binary and regressed within bootstrap-CI-width on three-class (−0.011), full-ordinal (−0.010), and NSD+ subgroup (−0.014). Paper 1's 22-feature schema is retained. Full tables, mechanism discussion, and reproducibility details in `supplementary_tabular_33feat_sensitivity.md`.

S-5. Confounder Sensitivity (Age, Sex, Enrollment Wave, DaT-SPECT Protocol)

Four pre-specified sensitivity analyses directly address whether the reported CatBoost binary AUC (0.979, Table I) is confounded by age, sex, PPMI enrollment wave, or DaT-SPECT acquisition protocol. **(A) Age-matched 1:1 sensitivity**: a greedy nearest-neighbour matched cohort (n=1,558; caliper ±2 yr≈0.2×SD per Austin 2011 [3]; residual matched Δ age=-0.002 yr) yields binary AUC 0.969 [95% CI 0.960–0.978] and three-class macro-AUC 0.931, both within one CI width of Table I. The small residual agrees with Schmitz-Steinkrüger *et al.* (2021), who reported that age and sex correction does not materially improve DaT-SPECT diagnostic performance because age and sex together explain $<10\,\%$ of SBR variance versus the $\sim 50\,\%$ reduction that defines pathological loss [44]. **(B) Sex-stratified**: male (n=849) and female (n=1,352) binary AUCs are identical (0.9770 vs. 0.9766; bootstrap Δ=+0.0004, 95% CI $[-0.013, +0.015]$, p=0.914). No significant sex interaction on any of the four targets. **(C) Enrollment-wave leave-one-cohort-out**: three PPMI waves (early 2010–2013, middle 2014–2020, late 2021–2025) yield binary AUC range 0.947–0.992 (mean 0.965 ± 0.024) and three-class macro-AUC range 0.930–0.946 (mean 0.939 ± 0.008), demonstrating cross-era generalisability; this stratification conflates scanner-era drift with cohort-recruitment-era shifts. **(D) DaT-SPECT protocol leave-one-cohort-out** (new): on the 2,137-patient analyzed-baseline-scan cohort, held-out binary AUCs are 0.967 [95% CI 0.952–0.979] (protocol 001, N=965) and 0.989 [0.982–0.995] (protocol 002, N=1,141), cross-protocol mean 0.978 ± 0.016; three-class macro-AUCs are 0.939 and 0.936 (mean 0.937 ± 0.002). The edge bucket (protocols 004 + T011, N=31) is retained in training but not independently held out. Analysis D partially disentangles the scanner/reconstruction-era effect from Analysis C's cohort-recruitment-era component. Uncontrolled confounders not addressed (DICOM-header scanner make/model / medication state at acquisition / comorbidities / handedness laterality) and the rationale for the enrollment-wave-over-site-LOSO substitution ('`site_aprv`' is a site-approval date, not a site identifier) are detailed in `supplementary_confounder_sensitivity.md`. Reproduction scripts (seed 42) are enumerated in the reproducibility package.

References

[1] M. Chen *et al.*, "AdaMedGraph: Adaboost-enhanced graph neural network for Parkinson's disease progression prediction," *npj Parkinson's Dis.*, 2024.

[2] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *arXiv:2107.07511*, 2021.

[3] P. C. Austin, "Optimal caliper widths for propensity-score matching when estimating differences in means and differences in proportions in observational studies," *Pharm. Stat.*, vol. 10, no. 2, pp. 150–161, 2011. doi: 10.1002/pst.433.

[4] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive inference with the jackknife+," *Ann. Statist.*, vol. 49, no. 1, pp. 486–507, 2021. doi: 10.1214/20-AOS1965.

[5] R. Bonnier, M. Kubler, F.-X. Standaert *et al.*, "Assessing the Robustness of Ordinal Classifiers against Imbalanced and Shifting Distributions," *Proceedings of Machine Learning Research*, vol. 183, 2022. DOI: proceedings.mlr.press/v183/bonnier22a

[6] H. Braak, K. Del Tredici, U. Rüb, R. A. I. de Vos, E. N. H. Jansen Steur, and E. Braak, "Staging of brain pathology related to sporadic Parkinson's disease," *Neurobiol. Aging*, vol. 24, no. 2, pp. 197–211, 2003.

[7] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[8] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery Data Mining*, 2016, pp. 785–794.

[9] J. Cohen, "Weighted kappa: nominal scale agreement with provision for scaled disagreement or partial credit," *Psychol. Bull.*, vol. 70, no. 4, pp. 213–220, 1968.

[10] G. S. Collins, K. G. M. Moons, P. Dhiman, *et al.*, "TRIPOD+AI statement: updated reporting guidelines for clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, Art. no. e078378, 2024.

[11] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.

[12] T. Dam, C. Gochanour, M. Brumm *et al.*, "Neuronal alpha-Synuclein Disease integrated staging system performance in PPMI, PASADENA, and SPARK baseline cohorts," *npj Parkinson's Disease*, vol. 10(1), 2024. DOI: 10.1038/s41531-024-00789-w

[13] A. Diaz-Rincon *et al.*, "Conformal prediction for Parkinson's disease medication needs," *arXiv preprint*, 2025.

[14] A. Bentivoglio *et al.*, "Clinical and imaging characteristics of Parkinson's disease with negative alpha-synuclein seed amplification assay," *Mov. Disord.*, 2026. doi: 10.1002/mds.70197.

[15] *et al.*, "New diagnostic and staging framework applied to established PD in the BioFIND cohort," *npj Parkinsons Dis.*, vol. 11, 2025. doi: 10.1038/s41531-025-00992-3.

[16] J.-E. Ding, C.-H. Chu, M.-N. L. Huang, C.-C. Hsu, "Dopamine Transporter SPECT Image Classification for Neurodegenerative Parkinsonism via Diffusion Maps and Machine Learning Classifiers," *Medical Image Understanding and Analysis (MIUA)*, vol. arXiv:2104.02066, 2021. DOI: arXiv:2104.02066

[17] J.-E. Ding, C.-C. Hsu, F. Liu, "Parkinson's Disease Classification Using Contrastive Graph Cross-View Learning with Multimodal Fusion of SPECT Images and Clinical Features," *arXiv preprint*, vol. arXiv:2311.14902, 2023. DOI: arXiv:2311.14902

[18] A. J. Espay *et al.*, "Clinical and biological characterization of patients with Parkinson's disease with and without improvement on dopaminergic medications," *Lancet Neurol.*, 2025.

[19] A. J. Espay *et al.*, "Refutation of the αSyn-SAA-based staging for Parkinson's progression (Neuronal α-Synuclein Disease-Integrated Staging System [NSD-ISS])," *Mov. Disord.*, 2025. doi: 10.1002/mds.30269.

[20] P. Eusebi *et al.*, "Diagnostic utility of [123I]FP-CIT SPECT in differentiating Parkinson's disease from essential tremor: a meta-analysis," *Eur. J. Nucl. Med. Mol. Imaging*, vol. 44, no. 3, pp. 495–514, 2017. doi: 10.1007/s00259-016-3597-9.

[21] B. Fischl, "FreeSurfer," *NeuroImage*, vol. 62, no. 2, pp. 774–781, 2012. doi: 10.1016/j.neuroimage.2012.01.021.

[22] Authors (GNN DILI) *et al.*, "Improving drug-induced liver injury prediction using graph neural networks with augmented graph features (nested-CV framework for GNN hyperparameter optimization)," *Journal of Cheminformatics*, vol. 17, 2025. DOI: 10.1186/s13321-025-01068-3

[23] C. G. Goetz *et al.*, "Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results," *Mov. Disord.*, vol. 23, no. 15, pp. 2129–2170, 2008.

[24] P. Gonzalez-Latapi, C. Gochanour, H. Cho *et al.*, "Eleven Years of Change: Disease Progression in Biomarker-Defined Sporadic Parkinson's Disease," *medRxiv preprint*, vol. , 2024. DOI: 10.1101/2024.10.09.24315191

[25] Y. Gorishniy, I. Rubachev, V. Khrulkov, A. Babenko, "Revisiting Deep Learning Models for Tabular Data," *NeurIPS*, vol. arXiv:2106.11959, 2021. DOI: arXiv:2106.11959

[26] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?" in *Proc. NeurIPS*, 2022.

[27] K. Huang, Y. Jin, E. Candes, and J. Leskovec, "Uncertainty quantification over graph with conformalized graph neural networks," in *Proc. NeurIPS*, 2023.

[28] G. Ke *et al.*, "LightGBM: a highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, 2017.

[29] S. Khosousi *et al.*, "Increased CSF DOPA Decarboxylase Correlates with Lower DaT-SPECT Binding: analyses in Biopark and PPMI cohorts," *Movement Disorders*, vol. 39(10):1881–1885, 2024. DOI: 10.1002/mds.29835

[30] K. Marek *et al.*, "The Parkinson's Progression Markers Initiative (PPMI) — establishing a PD biomarker cohort," *Ann. Clin. Transl. Neurol.*, vol. 5, no. 12, pp. 1460–1477, 2018.

[31] V. Taquet, V. Blot, T. Morzadec, L. Lacombe, and N. Brunel, "MAPIE: an open-source library for distribution-free uncertainty quantification," *arXiv:2207.12274*, 2022.

[32] B. Mollenhauer, C. J. Caspell-Garcia, C. T. Coffey, *et al.*, "Longitudinal CSF biomarkers in patients with early Parkinson disease and healthy controls," *Neurology*, vol. 89, no. 19, pp. 1959–1969, 2017. doi: 10.1212/WNL.0000000000004609.

[33] J. Muhammad *et al.*, "Trustworthy AI for medical decisions: Adversarially robust and fair machine learning prediction for Parkinson's disease," *PLOS ONE*, vol. 21(2):e0342062, 2026. DOI: 10.1371/journal.pone.0342062

[34] M. A. Nalls, C. Blauwendraat, C. L. Vallerga, *et al.*, "Identification of novel risk loci, causal insights, and heritable risk for Parkinson's disease: a meta-analysis of genome-wide association studies," *Lancet Neurol.*, vol. 18, no. 12, pp. 1091–1102, 2019. doi: 10.1016/S1474-4422(19)30320-5.

[35] R. B. Postuma *et al.*, "MDS clinical diagnostic criteria for Parkinson's disease," *Mov. Disord.*, vol. 30, no. 12, pp. 1591–1601, 2015.

[36] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Proc. NeurIPS*, 2018.

[37] J. Quan, L. Xu, R. Xu, T. Tong, J. Su, "DaTscan SPECT Image Classification for Parkinson's Disease," *arXiv preprint*, vol. arXiv:1909.04142, 2019. DOI: arXiv:1909.04142

[38] E. Rissanen *et al.*, "Machine learning applications for Parkinson's disease: a systematic review," *NPJ Digit. Med.*, 2022.

[39] *et al.*, "Reconsidering the clinical foundations of NSD-ISS in PD staging," *Mov. Disord.*, 2025. doi: 10.1002/mds.70061.

[40] M. J. Russo *et al.*, "Application of the NSD-ISS biological staging framework in the BioFIND cohort," *npj Parkinson's Dis.*, 2025.

[41] A. Sar, P. Puri, H. Naz *et al.*, "Multi-modal deep learning framework for early detection of Parkinson's disease using neurological and physiological data," *Scientific Reports*, vol. 15:1, 2025. DOI: 10.1038/s41598-025-21407-6

[42] S. Shokrpour, A. MoghadamFarid, S. Bazzaz Abkenar *et al.*, "Machine learning for Parkinson's disease: a comprehensive review of datasets, algorithms, and challenges," *npj Parkinson's Disease*, vol. 11:1, 2025. DOI: 10.1038/s41531-025-01025-9

[43] T. Simuni *et al.*, "Reply to: Refutation of the αSyn-SAA-based staging for Parkinson's progression," *Mov. Disord.*, 2025. doi: 10.1002/mds.30272.

[44] H. Schmitz-Steinkrüger, C. Lange, I. Apostolova, F. L. Mathies, L. Frings, S. Klutmann, S. Hellwig, P. T. Meyer, and R. Buchert, "Impact of age and sex correction on the diagnostic performance of dopamine transporter SPECT," *Eur. J. Nucl. Med. Mol. Imaging*, vol. 48, no. 5, pp. 1445–1459, May 2021. doi: 10.1007/s00259-020-05085-2.

[45] J. P. Seibyl, "DaT-SPECT in Parkinson's disease: clinical utility and limitations," *J. Nucl. Med.*, 2018.

[46] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *J. Mach. Learn. Res.*, vol. 9, pp. 371–421, 2008.

[47] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.

[48] T. Simuni *et al.*, "A biological definition of neuronal alpha-synuclein disease: towards an integrated staging system for research," *Lancet Neurol.*, vol. 23, no. 2, pp. 178–190, Feb. 2024.

[49] T. Simuni *et al.*, "Validation of the NSD-ISS in PPMI, PASADENA, and SPARK," *npj Parkinson's Dis.*, 2024.

[50] T. Simuni *et al.*, "Longitudinal clinical and biomarker characteristics of non-manifesting carriers and participants with Lewy body spectrum disease in the PPMI study using the NSD-ISS: A prospective cohort study," *Lancet Neurol.*, vol. 24, no. 2, pp. 127–142, Feb. 2025.

[51] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2018.

[52] A. Varrone *et al.*, "European multicentre database of healthy controls for [123I]FP-CIT SPECT (ENC-DAT): age-related effects, gender differences and evaluation of different methods of analysis," *Eur. J. Nucl. Med. Mol. Imaging*, vol. 40, no. 2, pp. 213–227, 2013. doi: 10.1007/s00259-012-2276-8.

[53] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, 2nd ed. Springer, 2022.

[54] N. Wakasugi, H. Takano, M. Abe *et al.*, "Harmonizing multi-site data with the ComBat method for enhanced PD diagnosis

via DAT-SPECT," *Front. Neurol.*, vol. 15, Art. no. 1306546, 2024. doi: 10.3389/fneur.2024.1306546.
[55] H.-J. Ye, S.-Y. Liu, H.-R. Cai, Q.-L. Zhou, D.-C. Zhan, "A Closer Look at Deep Learning Methods on Tabular Datasets," *arXiv preprint*, vol. arXiv:2407.00956, 2024. DOI: arXiv:2407.00956
[56] G. Zabërgja, A. Kadra, J. Grabocka *et al.*, "Tabular Data: Is Deep Learning All You Need?," *arXiv preprint*, vol. arXiv:2402.03970, 2024. DOI: arXiv:2402.03970
[57] Z. Zhang, X. Chen, Y. Shi *et al.*, "Provably Minimum-Length Conformal Prediction Sets for Ordinal Classification," *arXiv preprint*, vol. arXiv:2511.16845, 2025. DOI: arXiv:2511.16845
[58] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *J. R. Stat. Soc. B*, vol. 67, no. 2, pp. 301–320, 2005.
[59] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, pp. 319–326, 2025.
[60] N. Erickson, J. Mueller, A. Shirkov, H. Zhang, P. Larroy, M. Li, and A. Smola, "AutoGluon-Tabular: Robust and accurate AutoML for structured data," *arXiv preprint* arXiv:2003.06505, 2020.
[61] T. Shadbahr, M. Roberts, J. Stanczuk, J. Gilbey, P. Teare, S. Dittmer, M. Thorpe, R. V. Torne, E. Sala, P. Liò, M. Patel, AIX-COVNET Collaboration, J. A. D. Aston, C.-B. Schönlieb, and J. H. F. Rudd, "The impact of imputation quality on machine learning classifiers for datasets with missing values," *Commun. Med.*, vol. 3, no. 1, p. 139, 2023.
[62] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *J. Mach. Learn. Res.*, vol. 11, pp. 2079–2107, 2010.
[63] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1321–1330.
[64] M. P. Naeini, G. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. AAAI Conf. Artif. Intell.*, 2015, pp. 2901–2907.
[65] P. C. Austin and E. W. Steyerberg, "Graphical assessment of internal and external calibration of logistic regression models by using loess smoothers," *Stat. Med.*, vol. 39, no. 6, pp. 727–746, 2020.
[66] M. Sadinle, J. Lei, and L. Wasserman, "Least ambiguous set-valued classifiers with bounded error levels," *J. Amer. Stat. Assoc.*, vol. 114, no. 525, pp. 223–234, 2019.
[67] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *J. Pharmacokinet. Biopharm.*, vol. 15, no. 6, pp. 657–680, 1987.
[68] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017.
[69] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Comput.*, vol. 14, no. 1, pp. 21–41, 2002.
[70] H. Boström and U. Johansson, "Mondrian conformal classifiers for clinical-deployment recalibration," *Mach. Learn.*, vol. 114, no. 3, pp. 1217–1248, 2025.